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SELFLESS C^*-CORRESPONDENCES, OPERATOR VALUED C^*-PROBABILITY SPACES AND COMPLETELY POSITIVE MAPS

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Abstract. We develop a general theory of selflessness for C^*-correspondences, with several applications. Specializing this theory to completely positive maps, in particular to conditional expectations, gives rise to a novel notion of relative selflessness. Among applications, the machinery developed here yields new examples of selfless C^*-algebras, for instance among minimal tensor products and reduced crossed products, new examples of MF C^*-algebras arising as reduced amalgamated free products, a conceptually new proof of Kirchberg's $\mathcal{O}_\infty$-absorption theorem, and the lack of "phantom" traces on ultrapowers.

1. Introduction

The notion of selfless C^*-probability space was introduced by the fifth author in [29]. Recall that a C*-probability space (A, ρ), where A is a unital C*-algebra and ρ is a state on A inducing a faithful GNS representation, is called selfless if the embedding

$$(A, \rho) \to (A, \rho) * (C, \kappa)$$

is existential for some C^*-probability space (C, κ), with $C \neq \mathbb{C}$ and κ a state inducing a faithful GNS representation of C. As demonstrated in [29], selfless C^*-algebras possess many desirable properties, including simplicity, having at most one trace, and in the tracial case having stable rank 1 and the strict comparison property.

The work [1] proved selflessness for a wide class of reduced group C^*-algebras, using an approach involving the rapid decay property. Selflessness has since been established for broad classes of reduced group C^*-algebras [22, 25, 38, 3], twisted group C^*-algebras [27, 9], reduced free products [16, 15, 8], graph products [8], amalgamated free products and HNN extensions [10], linear groups [35], and compact quantum groups [15]. Notably, in [25] Ozawa settled several natural questions concerning selflessness, developed a streamlined approach to proving this property, defined a notion of complete selflessness, and introduced the PHP property for discrete groups.

This article presents a new framework that expands the reach of selflessness to a very broad generality, with various applications. We begin by discussing the case of arbitrary completely positive maps. Let A be a unital C*-algebra. Denote by $CP(A)$ the set of cp maps from A to A. To formulate selflessness of $\phi \in CP(A)$, we rely on Shlyakhtenko's construction of the C*-algebra $C^*(A, s_\phi)$ generated by A and a semicircular element s_ϕ with covariance ϕ. We call $\phi \in CP(A)$ selfless if

the canonical embedding

$$(A;\phi) \to (C^*(A,s_\phi);\phi\circ E_\Omega)$$

is positively existential. This means that for some ultrapower $A^{\mathcal{U}}$ there is a *-homomorphism $\sigma\colon C^*(A,s_\phi)\to A^{\mathcal{U}}$ such that $\sigma|_A$ induces the diagonal embedding and σ intertwines the cp maps $\phi\circ E_\Omega$ and $\phi^{\mathcal{U}}$.

The case when the cp map is a conditional expectation is of special interest, and further illuminates selflessness in this new setting. Let $E\colon A\to B$ be a conditional expectation (onto a unital C*-subalgebra B) inducing a faithful GNS representation of A. Then E is selfless if and only if there exists a C*-probability space (C,κ), with $C\neq\mathbb{C}$ and κ inducing a faithful GNS representation of C, for which the embedding

$$(A;E)\to(A;E)*_B(B\otimes C;\mathrm{id}_B\otimes\kappa)$$

is existential (where the right-hand side is an amalgamated reduced free product, and the tensor product is the minimal one). Thus, there exists a *-homomorphism $\sigma\colon A*_B(B\otimes C)\to A^{\mathcal{U}}$ whose restriction to A agrees with the diagonal inclusion and that intertwines the expectations $E*_B(\mathrm{id}_B\otimes\kappa)$ and $E^{\mathcal{U}}$. Note that, besides the free independence relation that this enforces in $A^{\mathcal{U}}$—$\sigma(B\otimes C)$ is free from A with amalgamation over B—selflessness of E enforces a commutation relation: $\sigma(C)\subseteq B'\cap A^{\mathcal{U}}$. By choosing E to be a state (i.e., $B=\mathbb{C}$), we recover the notion of selfless C*-probability space.

To investigate selfless cp maps and expectations, we go one step further in generality and define selflessness for a C*-correspondence H (over A) endowed with a real structure $K\subseteq H$. To formulate selflessness for the pair (H,K), we again appeal to Shlyakhtenko's C*-algebra generated by A and the A-valued semicircular system associated to the pair (H,K) [33]. Selfless cp maps are then precisely those for which the C*-correspondence $A\otimes_\phi A$, obtained via the KSGNS construction, endowed with the real structure generated by $1\otimes 1$, is selfless. In this introduction, we will confine our discussion to the settings of cp maps and conditional expectations, as this allows for a less technical presentation.

We also work with a notion of weak selflessness, which, as its name indicates, is implied by selflessness. A cp map $\phi\in CP(A)$ is called weakly selfless if the embedding

$$A\to C^*(A,s_\phi)*_A C^*(A,s_\phi)*_A\cdots$$

is a positively existential embedding of C*-algebras, where the right-hand side is the infinite amalgamated reduced free product of $(C^*(A,s_\phi),E_\Omega)$ with itself. Note that we do not enforce an intertwining of the associated cp maps. Weak selflessness behaves more flexibly than selflessness (e.g., the identity map $\mathrm{id}_A\colon A\to A$ can never be selfless but may be weakly selfless), but is strong enough to have structural implications for the underlying C*-algebra.

In the state case, if (A,ρ) is selfless then it has the Dixmier property. This in turn implies that A is simple, and that it is either traceless or has at most one trace (if ρ is tracial). In the setting of cp maps, we have the following:

Proposition 1.1. *Let A be a unital C*-algebra.*

(i) If $\phi\in CP(A)$ is weakly selfless, then it is approximately selfadjoint inner, that is to say, it is a point-norm limit of cp maps of the form

$$x\mapsto\sum_{i=1}^n h_i x h_i,$$

where $h_i \in A$ is selfadjoint.

(ii) If $E\colon A \to B$ is a weakly selfless conditional inducing a faithful representation of A, then $\Delta_A \circ E\colon A \to A^{\mathcal{U}}$ is a point norm limit of maps of the form

$$x \mapsto \frac{1}{n}\sum_{i=1}^{n} u_i^* x u_i,$$

where $u_i \in B' \cap A^{\mathcal{U}}$ is a unitary for all i. (Here $\mathcal{U}$ is an ultrafilter on $\mathbb{N}$ and Δ_A the diagonal inclusion of A in its ultrapower $A^{\mathcal{U}}$.)

As a consequence, if ϕ is weakly selfless then it preserves closed two-sided ideals of A, and if it is also unital, then if preserves the tracial states of A.

In the state case, a strong dichotomy holds for selfless C*-probability spaces: their underlying C*-algebras are either purely infinite or have stable rank one [29], [12]. Moreover, in the tracial state case, the C*-algebra has strict comparison of positive elements by its unique trace.

Although the purely infinite/stable-rank-one dichotomy does not persist for general cp maps, weak selflessness continues to impose regularity on the underlying algebra. We show that if a unital C^*-algebra A admits a weakly selfless unital cp map, then every bounded trace on $A^{\mathcal{U}}$ is essentially a limit trace (i.e., $A^{\mathcal{U}}$ has no "phantom traces"). The interactions between regularity properties and selflessness of expectations, cp maps, and C*-correspondences, will be the subject of future research.

To isolate the purely infinite case, we introduce the notion of Toeplitz selflessness. We call (A, ϕ) Toeplitz selfless if the embedding

$$(A; \phi) \to (C^*(A, \ell_\phi); \phi \circ E_\Omega)$$

is positively existential, where this time $C^*(A, \ell_\phi)$ is the Toeplitz-Pimsner C*-algebra associated to the KSGNS C*-correspondence $A \otimes_\phi A$, and ℓ_ϕ is the left creation operator. In the case of an expectation E inducing a faithful GNS representation of A, Toeplitz selflessness is equivalent to simply asking that the embedding

$$(A; E) \to (A; E) *_B (B \otimes C; \mathrm{id}_B \otimes \kappa)$$

is existential for some C and a *non-tracial* GNS-faithful state κ. Toeplitz selfless cp maps are *completely* selfless.

Theorem 1.2. *If a unital C*-algebra A admits a weakly Toeplitz selfless cp map $\phi \in CP(A)$ such that* $\mathrm{Ideal}(\phi(a)) = \mathrm{Ideal}(a)$ *for all $a \in A_+$, then A is strongly purely infinite (in the sense of Kirchberg and Rørdam* [21]*).*

A substantial part of the paper is devoted to permanence results for the new notions of selflessness. We mention some here:

Theorem 1.3. *Let A be a unital C*-algebra.*

(i) If $\phi_n \in CP(A)$ for $n = 1, \ldots$ are weakly selfless maps, and $\phi_n \to \phi$ in point norm, then ϕ is weakly selfless.

(ii) If $\phi, \psi \in CP(A)$ commute, ϕ is selfless, and ψ is approximately selfadjoint inner, then $\psi \circ \phi$ is selfless.

(iii) If $\phi \in CP(A)$ is selfless, $\psi \in CP(B)$ is approximately selfadjoint inner, and B is exact, then $\phi \otimes \psi$ is selfless. If ϕ is completely selfless, then same conclusion holds without the assumption that B is exact.

(iv) If $E_1 \colon A_1 \to B$ is a selfless conditional expectation, and $E_2 \colon A_2 \to B$ is a conditional expectation inducing a faithful GNS representation, then $E_1 *_B E_2 \colon A_1 *_B A_2 \to B$ is selfless.

In [25], Ozawa shows that a tensor product of exact selfless C*-algebras is selfless. Theorem 1.3 (iii) extends this result to cp maps, while also relaxing the assumption on one of the cp maps to be approximately selfadjoint inner. Specializing to states, this shows that if (A, ρ) is selfless, and B is exact, unital, simple, with unique trace τ, then $(A \otimes B, \rho \otimes \tau)$ is selfless. This yields several new examples of C^*-selfless groups, including $G \times H$ where G has Ozawa's PHP property (such as C^*-simple acylindrically hyperbolic or linear groups) and H is C^*-simple.

A natural setting to investigate selflessness of expectations is the reduced crossed products. By an adaptation of Ozawa's PHP technique we prove the following result:

Theorem 1.4. *If a group G with the PHP property acts on a unital C^*-algebra A by approximately inner automorphisms, then the conditional expectation $E \colon A \rtimes_r G \to A$ is selfless.*

Combined with the previous permanence results, this yields new examples of selfless C*-algebras. For example, since every automorphism of the Jiang-Su algebra $\mathcal{Z}$ is approximately inner, the expectation $E \colon \mathcal{Z} \rtimes_r G \to \mathcal{Z}$ is selfless for any action of a PHP group G on $\mathcal{Z}$. Further, since $\mathcal{Z}$ is simple with unique trace τ, it follows by Theorem 1.3 (ii) that the trace $\tau \circ E \colon \mathcal{Z} \rtimes_r G \to \mathbb{C}$ is selfless, i.e., $(\mathcal{Z} \rtimes_r G, \tau \circ E)$ is a selfless C*-probability space. This notably yields new examples of C^*-algebras with strict comparison.

Our final group of results concerns the interaction between selflessness and nuclearity. Here selflessness becomes a criterion for tensorial absorption. Indeed, the purely infinite/unique trace-strict comparison dichotomy, together with the Kirchberg and Matui–Sato absorption theorems, implies that every nuclear selfless C^*-probability space has $\mathcal{Z}$-stable C*-algebra.

A direct link between selflessness of cp maps and tensorial stability is that if A is separable and the identity map id_A is weakly selfless, then A is $\mathcal{Z}$-stable, while if A is separable and id_A is weakly Toeplitz selfless, then A is $\mathcal{O}_\infty$-stable. We deduce from this and the permanence results on selflessness the following:

Theorem 1.5. *Let A be a separable unital C^*-algebra and $\phi \in CP(A)$.*

(i) If ϕ is both weakly selfless and real relatively nuclear (Definition 8.1), then A is $\mathcal{Z}$-stable.

(ii) If ϕ is both weakly Toeplitz selfless and relatively nuclear (Definition 8.1), then A is $\mathcal{O}_\infty$-stable.

The proof of Ozawa's [25, Theorem 3] shows that if A is simple, unital, and purely infinite, then (A, ρ) is Toeplitz selfless (for any state ρ). Combined with (ii) of the previous theorem, this gives a new proof of Kirchberg's $\mathcal{O}_\infty$-stability theorem ([30]): if A is separable, purely infinite, simple, unital, and nuclear, then $A \otimes \mathcal{O}_\infty \cong A$.

As another application of our results on selflessness and nuclearity, we establish a special case of a conjecture of Hayes [31, Conjecture 1.2]. The conjecture asserts that if G_1 and G_2 are MF groups with a common amenable subgroup H, then the amalgamated free product $G_1 *_H G_2$ is MF. We prove that this is the case if $G_1 = G_2$: if G is an MF group and $H \leq G$ is amenable, then $G *_H G$ is MF.

Organization of the paper. The paper is organized as follows: In Sections 3 and 4 we develop the general framework of selflessness for C*-correspondences. In Section 5 we consider selfless cp maps. In Section 6 we obtain some regularity properties for the underlying C*-algebra of a weakly selfless cp map. In Section 7 we specialize our results to conditional expectations. In Section 8 we investigate the interaction between selflessness and nuclearity. In Section 9 we consider expectations arising from crossed products.

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2. Preliminaries

Throughout, all C*-algebras are assumed unital and all *-homomorphisms are unital.

2.1. **C*-correspondences.** By an A-B C*-correspondence we mean a right Hilbert B-module H together with a unital *-homomorphism $A \to B(H)$, which we write as a left action. (We denote by $B(H)$ the C*-algebra of adjointable operators on H.)

We review some facts about C*-correspondences here. For more background, see [23, Chapter 4] and [5].

We call an A-A C*-correspondence H a C*-correspondence over A. Note that A is a C*-correspondence over itself with the obvious left and right actions and standard Hilbert C*-module inner product $\langle a, b\rangle = a^*b$.

A covariant map $(\Phi, \sigma)\colon H_1 \to H_2$ between C*-correspondences respectively over A and B consists of a *-homomorphism $\sigma\colon A \to B$ and a linear map $\Phi\colon H_1 \to H_2$ such that

(1) $\Phi(avb) = \sigma(a)\Phi(v)\sigma(b)$ for all $a, b \in A$ and $v \in H_1$,
(2) $\sigma(\langle v, w\rangle) = \langle \Phi(v), \Phi(w)\rangle$ for all $v, w \in H_1$.

We often emphasize Φ and refer to σ as the underlying *-homomorphism. A covariant map $H \to B$ into the identity C*-correspondence of a C*-algebra B is called a representation of H.

We often use the following lemma:

Lemma 2.1. *Let $\sigma\colon A \to C$ be an embedding of C*-algebras. Let H and H' be C*-correspondences over A and C, respectively. Suppose that H is generated (as a closed bimodule) by an indexed collection of vectors $(v_i)_{i\in I}$, and that we have an assignment $v_i \mapsto w_i \in H'$ such that*

$$\sigma(\langle v_i, av_j\rangle) = \langle w_i, \sigma(a)w_j\rangle$$

*for all $a \in A$ and all $i, j \in I$. Then there is a unique covariant embedding $H \to H'$ such that $v_i \mapsto w_i$ and with underlying *-homomorphism σ.*

Given A-B and B-C C*-correspondences H_1 and H_2, $H_1 \otimes_B H_2$ is obtained by "separation and completion" from their algebraic tensor product with semi-inner product defined as

$$\langle v_1 \otimes w_1, v_2 \otimes w_2 \rangle = \langle w_1, \langle v_1, v_2 \rangle w_2 \rangle.$$

2.2. **The Toeplitz-Pimsner C*-algebra.** We recall the notation for the Toeplitz-Pimsner algebra associated to a C*-correspondence. For background, see [20, Proposition 3.3] and [5, Section 4.6].

Let H be a C*-correspondence over A. Its Fock C*-correspondence is

$$\mathcal{F}(H) = A \oplus \bigoplus_{n \geq 1} H^{\otimes_A n}.$$

For $v \in H$, let $\ell_v \in B(\mathcal{F}(H))$ denote the left creation operator. Together with the left action of A on $\mathcal{F}(H)$, the assignment $v \mapsto \ell_v$ gives a representation of H on $\mathcal{F}(H)$. We denote by

$$\mathcal{T}(H) = C^*(A, \{\ell_v : v \in H\})$$

the Toeplitz-Pimsner C*-algebra of H. We denote by $E_\Omega \colon \mathcal{T}(H) \to A$ the vacuum expectation, obtained by compressing to the A-summand:

$$E_\Omega(x) = P_0 x P_0.$$

The representation of H in $\mathcal{T}(H)$ has the following universal property: if (Φ, σ) is a representation of H in a C*-algebra D, then there is a *-homomorphism $\pi \colon \mathcal{T}(H) \to D$ such that

$$\pi(\ell_v) = \Phi(v), \qquad \pi(a) = \sigma(a)$$

for all $v \in H$ and $a \in A$. See [20, Proposition 6.5], [5, Theorem 4.6.18].

We will use the following application of the functoriality of Toeplitz-Pimsner C*-algebra construction, which we informally refer to as "Toeplitz exactness"; see [11].

Lemma 2.2 (Toeplitz exactness)**.** *Let $\mathcal{U}$ be an ultrafilter on I. Let H be a C*-correspondence over A, and let H_i be a C*-correspondence over A_i, for each $i \in I$. Suppose that there is a covariant map*

$$(\Phi, \sigma) \colon H \longrightarrow \prod_{\mathcal{U}} H_i,$$

*where $\sigma \colon A \to \prod_{\mathcal{U}} A_i$ is the underlying *-homomorphism. Then there exists a *-homomorphism*

$$\widetilde{\sigma} \colon \mathcal{T}(H) \longrightarrow \prod_{\mathcal{U}} \mathcal{T}(H_i)$$

extending σ and satisfying $\widetilde{\sigma}(\ell_v) = [(\ell_{v_i})_i]$ whenever $\Phi(v) = [(v_i)_i]$. Moreover,

$$E_{\mathcal{U}} \circ \widetilde{\sigma} = \sigma \circ E_H,$$

where $E_{\mathcal{U}} \colon \prod_{\mathcal{U}} \mathcal{T}(H_i) \to \prod_{\mathcal{U}} A_i$ is given by $E_{\mathcal{U}}([(x_i)_i]) = [(E_{H_i}(x_i))_i]$.

*Suppose, in addition, that H and the H_i are equipped with real structures K and K_i, respectively, and that $\Phi(K) \subseteq \prod_{\mathcal{U}} K_i$. Then $\widetilde{\sigma}$ restricts to a *-homomorphism*

$$\mathcal{S}(H, K) \longrightarrow \prod_{\mathcal{U}} \mathcal{S}(H_i, K_i)$$

intertwining the corresponding expectations. (For definition of real structures and $\mathcal{S}$, see Section 2.3.)

Proof. See also [11]. For each $i \in I$, let $L_i \colon H_i \to \mathcal{T}(H_i)$ be the canonical representation, $v \mapsto \ell_v$. The map

$$\prod_{\mathcal{U}} H_i \longrightarrow \prod_{\mathcal{U}} \mathcal{T}(H_i), \qquad [(v_i)_i] \longmapsto [(\ell_{v_i})_i],$$

together with the canonical *-homomorphism

$$\prod_{\mathcal{U}} A_i \longrightarrow \prod_{\mathcal{U}} \mathcal{T}(H_i),$$

is a representation of $\prod_{\mathcal{U}} H_i$. Precomposing this representation with (Φ, σ) gives a representation of H in $\prod_{\mathcal{U}} \mathcal{T}(H_i)$. The universal property of $\mathcal{T}(H)$ now gives $\widetilde{\sigma}$.

The identity

$$E_{\mathcal{U}} \circ \widetilde{\sigma} = \sigma \circ E_H$$

is verified on the dense linear span of the elements $\ell_\xi \ell_\eta^*$, where ξ and η range over tensor powers of H. Finally, if $v \in K$ and $\Phi(v) = [(v_i)_i]$ with $v_i \in K_i$, then

$$\widetilde{\sigma}(s_v) = [(s_{v_i})_i].$$

This proves the final assertion. □

2.3. **Shlyakhtenko's C*-algebra.** Let H be a correspondence over A. A norm closed real subspace $K \subseteq H$ is called a *real structure* if

(1) K generates H as closed A-A bimodule,
(2) K is closed under $v \mapsto a^* v a$ for all $a \in A$.

For the identity correspondence, one such real structure is the subspace of selfadjoint elements A_{sa}. If we assume that A is unital (as we shall always do), this is the real structure generated by the unit. We call the pair (H, K) a C*-correspondence with real structure. We sometimes write (H, K, A) to indicate the underlying C*-algebra and remove ambiguity.

For covariant maps between C*-correspondences with real structures (possibly over different C*-algebras) $(H_1, K_1) \to (H_2, K_2)$ we require that K_1 is mapped to K_2.

Let us now review Shlyakhtenko's C*-algebra of semicirculars associated to a C*-correspondence with real structure (H, K) [33]. The elements $s_v = \ell_v + \ell_v^* \in \mathcal{T}(H)$ are called semicircular elements. Given (H, K), we form the C*-algebra

$$\mathcal{S}(H, K) := C^*(A, \{s_v : v \in K\}) \subseteq \mathcal{T}(H),$$

which we call the C*-algebra of A-valued semicirculars associated to (H, K). We note that it comes equipped with the vacuum expectation onto A.

Lemma 2.3. *If $\Sigma \subseteq K$ generates K as a closed Jordan bimodule, then*

$$\mathcal{S}(H, K) = C^*(A, \{s_v : v \in \Sigma\}).$$

Proof. This follows from the identities $s_{rv+w} = r s_v + s_w$, $s_{ava^*} = a s_v a^*$, for $r \in \mathbb{R}$, $v, w \in K$, $a \in A$, and from the continuity of $v \mapsto s_v$. □

2.4. **Indexed families of cb maps.** We sometimes work with pairs $(A;(\phi_i)_{i\in I})$ where A is a unital C*-algebra and $(\phi_i)_{i\in I}$ is an indexed family of completely bounded (cb) maps $\phi_i \in \mathrm{CB}(A)$. By a morphism $\sigma\colon (A;(\phi_i)_{i\in I}) \to (B;(\psi_i)_{i\in I})$ between such objects we understand a *-homomorphism $\sigma\colon A \to B$ that intertwines the cb maps: $\psi_i\sigma = \sigma\phi_i$ for all i (notice that the index set is the same on both domain and codomain).

By the ultrapower of $(A;(\phi_i)_{i\in I})$ we understand $(A^{\mathcal{U}};(\phi_i^{\mathcal{U}})_{i\in I})$. We denote by Δ_A the diagonal embedding $(A;(\phi_i)_i) \to (A^{\mathcal{U}};(\phi_i^{\mathcal{U}})_{i\in I})$.

Definition 2.4. *Consider an embedding of C*-algebras with indexed families of cb maps:*

$$\theta\colon (A;(\phi_i)_{i\in I}) \to (B;(\psi_i)_{i\in I}).$$

*We say that θ is positively existential if there exists an ultrafilter $\mathcal{U}$ and a *-homomorphism*

$$\sigma\colon (B;(\psi_i)_{i\in I}) \to (A^{\mathcal{U}};(\phi_i^{\mathcal{U}})_{i\in I})$$

such that $\sigma\theta = \Delta_A$.

If σ may be chosen to also be an embedding, then we say that θ is existential.

Following [25, Section 6]*, we say that θ is completely positively existential if σ may be chosen such that for every C*-algebra C the induced *-homomorphism $\tilde{\sigma}\colon B\otimes C \to (A\otimes C)^{\mathcal{U}}$ is well defined (where the tensor products are minimal). If σ is additionally an embedding, we say that θ is completely existential.*

3. Selfless C*-correspondences

Let (H,K) be a C^*-correspondence over A with a real structure. Denote by $\mathcal{S}(H,K)$ Shlyakhtenko's C*-algebra and by $E\colon \mathcal{S}(H,K) \to A$ the vacuum conditional expectation.

For $v,w \in H$, set

$$\kappa_{v,w}(a) = \langle v, aw\rangle, \qquad a \in A.$$

Definition 3.1. *We call (H,K) selfless if*

$$(A;\{\kappa_{v,w} : v,w\in H\}) \to (\mathcal{S}(H,K);\{\kappa_{v,w}\circ E : v,w\in H\}) \tag{1}$$

is a positively existential embedding of C-algebras with indexed collections of cb maps. Equivalently, there exists $\sigma\colon \mathcal{S}(H,K) \to A^{\mathcal{U}}$ whose restriction to A agrees with the diagonal embedding Δ_A and such that*

$$\kappa_{v,w}^{\mathcal{U}}\circ\sigma = \Delta_A\circ\kappa_{v,w}\circ E \qquad \text{for all } v,w\in H. \tag{2}$$

If the embedding (1) *is completely positively existential, then we call (H,K) completely selfless.*

Remark 3.2. We prove all results for selflessness; the completely selfless versions are obtained by the same arguments.

In the next three lemmas we clarify some points concerning this definition. We fix a C*-correspondence with real structure (H,K), set $C = \mathcal{S}(H,K)$, and denote by $E\colon C \to A$ the vacuum expectation.

Lemma 3.3. *If the left action of A on H is faithful, and a *-homomorphism $\sigma\colon C \to A^{\mathcal{U}}$ satisfies* (2)*, then σ is an embedding. Thus, in this case the embedding* (1) *is existential.*

Proof. Suppose that $\sigma(c) = 0$ for some $c \in C$. Then

$$\Delta_A(\kappa_{v,w}(E(xcy))) = \kappa^{\mathcal{U}}_{v,w}(\sigma(xcy)) = 0$$

for all $x, y \in C$ and $v, w \in H$. By faithfulness of the left action of A on H, this implies that $E(xcy) = 0$ for all $x, y \in C$. Since E has faithful GNS, we get $c = 0$. □

Lemma 3.4. *Let $\sigma\colon C \to A^{\mathcal{U}}$ be a *-homomorphism such that $\sigma|_A = \Delta_A$. If $\Sigma \subseteq H$ generates H as a closed bimodule, then σ satisfies (2) if and only if*

$$\kappa^{\mathcal{U}}_{v,w} \circ \sigma = \Delta_A \circ \kappa_{v,w} \circ E \text{ for all } v, w \in \Sigma.$$

Thus, to conclude selflessness of (H, K), it is sufficient to check that the embedding

$$(A; \{\kappa_{v,w} : v, w \in \Sigma\}) \to (C; \{\kappa_{v,w} \circ E : v, w \in \Sigma\})$$

is positively existential.

Proof. The forward implication is immediate. For the converse, consider the set of all cb maps $\kappa \in CB(A)$ such that

$$\kappa^{\mathcal{U}} \circ \sigma = \Delta_A \circ \kappa \circ E.$$

It is straightforward to check that this set is closed under linear combinations, operator-norm limits, pre-composition by the left and right multiplication maps

$$L_a(x) = ax, \qquad R_a(x) = xa,$$

and post-compositions by arbitrary cb maps on A. We now use that

$$\kappa_{v_1+v_2,w} = \kappa_{v_1,w} + \kappa_{v_2,w}, \quad \kappa_{b_1 v b_2,w} = L_{b_2^*} \circ \kappa_{v,w} \circ L_{b_1^*},$$
$$\kappa_{v,w_1+w_2} = \kappa_{v,w_1} + \kappa_{v,w_2}, \qquad \kappa_{v,b_1 w b_2} = R_{b_2} \circ \kappa_{v,w} \circ R_{b_1},$$

to deduce that (2) holds for v, w ranging on a dense subset of H. Further, by the norm continuity of $(v, w) \mapsto \kappa_{v,w}$, we obtain that (2) holds for all $v, w \in H$. □

Consider the C*-correspondence over C induced by H:

$$H_C = L^2(C, E) \otimes_A H \otimes_A C.$$

Endow H_C with the real structure K_C generated by the set $1 \otimes K \otimes 1$. We note, for later use, that from the fact that K generates H as a closed bimodule, we deduce that the vacuum vector is cyclic for the action of C on $\mathcal{F}(H)$. Hence,

$$L^2(C, E) \cong \mathcal{F}(H) \tag{3}$$

as C-A C*-correspondences.

We have a covariant embedding $(H, K, A) \to (H_C, K_C, C)$ where $\Phi\colon H \to H_C$ is defined as

$$\Phi(v) = 1 \otimes v \otimes 1$$

and A is canonically embedded in C.

Lemma 3.5. *The pair (H, K) is selfless if and only if the covariant embedding $(H, K, A) \to (H_C, K_C, C)$ is a positively existential embedding of C*-correspondences with real structures. That is, for some ultrafilter $\mathcal{U}$ there exists a covariant map*

$$(\Psi, \sigma)\colon (H_C, K_C, C) \to (H^{\mathcal{U}}, K^{\mathcal{U}}, A^{\mathcal{U}})$$

such that

$$\Psi\Phi = \Delta_H, \quad \sigma|_A = \Delta_A.$$

Proof. Suppose that (H, K) is selfless and let $\sigma\colon C \to A^{\mathcal{U}}$ be a *-homomorphism witnessing this. Define $\Psi\colon H_C \to H^{\mathcal{U}}$ on elementary tensors $x \otimes v \otimes y \in C \otimes_A H \otimes_A C$ by

$$\Psi(x \otimes v \otimes y) = \sigma(x)\Delta_H(v)\sigma(y)$$

and extend it by linearity to their linear span. The calculation

$$\begin{aligned}\langle \Psi(x \otimes v \otimes y), \Psi(x' \otimes w \otimes y')\rangle &= \sigma(y)^*\langle \Delta_H(v), \sigma(x^*x')\Delta_H(w)\rangle\sigma(y') \\ &= \sigma(y)^*\Delta_A(\langle v, E(x^*x')w\rangle)\sigma(y') \\ &= \sigma(y^*\langle v, E(x^*x')w\rangle y') \\ &= \sigma(\langle x \otimes v \otimes y, x' \otimes w \otimes y'\rangle)\end{aligned}$$

shows both that Ψ is well defined and that it extends to a linear map from H_C to $H^{\mathcal{U}}$. It is immediate from its definition that (Ψ, σ) is covariant, that it sends K_C into $K^{\mathcal{U}}$, and that $\Psi \circ \Phi = \Delta_H$. Thus, the embedding $(H, K, A) \to (H_C, K_C, C)$ is positively existential.

Conversely, suppose that the embedding $(H, K, A) \to (H_C, K_C, C)$ is positively existential. Let $\sigma\colon C \to A^{\mathcal{U}}$ and $\Psi\colon H_C \to H^{\mathcal{U}}$ witness this. Then $\sigma|_A = \Delta_A$ and $\Psi \circ \Phi = \Delta_H$. For $c \in C$ and $v, w \in H$, the covariance of (Ψ, σ) gives

$$\begin{aligned}\kappa^{\mathcal{U}}_{v,w}(\sigma(c)) &= \langle \Delta_H(v), \sigma(c)\Delta_H(w)\rangle \\ &= \langle \Psi(1 \otimes v \otimes 1), \Psi(c \otimes w \otimes 1)\rangle \\ &= \Delta_A\big(\langle 1 \otimes v \otimes 1, c \otimes w \otimes 1\rangle\big) \\ &= \Delta_A(\langle v, E(c)w\rangle) \\ &= \Delta_A(\kappa_{v,w}(E(c))).\end{aligned}$$

Thus σ satisfies (2), and so (H, K) is selfless. □

Theorem 3.6. *If $(H_1, K_1) \subseteq (H_2, K_2)$ and (H_2, K_2) is (completely) selfless, then (H_1, K_1) is (completely) selfless. If $\{(H_i, K_i)\}_{i\in I}$ is an upward directed family of (completely) selfless A-A C*-subcorrespondences of (H, K) with dense union in (H, K), then (H, K) is also selfless.*

Proof. Suppose that $(H_1, K_1) \subseteq (H_2, K_2)$, with (H_2, K_2) selfless. The inclusion $H_1 \subseteq H_2$ induces a natural embedding $\mathcal{F}(H_1) \hookrightarrow \mathcal{F}(H_2)$. Identify $\mathcal{F}(H_1)$ with a sub-C*-correspondence of $\mathcal{F}(H_2)$ in this manner. Since $\mathcal{F}(H_1)$ is invariant under left creation operators $\ell_v \in \mathcal{T}(H_2)$ such that $v \in H_1$ and under their adjoints, the restriction $x \mapsto x|_{\mathcal{F}(H_1)}$ yields the isomorphism

$$C^*(A, \{\ell_v \in \mathcal{T}(H_2) : v \in H_1\}) \cong \mathcal{T}(H_1).$$

Identify $\mathcal{T}(H_1)$ with a C*-subalgebra of $\mathcal{T}(H_2)$ in this manner. Then, since $K_1 \subseteq K_2$, we obtain

$$\mathcal{S}(H_1, K_1) = C^*(A, \{s_v \in \mathcal{S}(H_2, K_2) : v \in K_1\}).$$

Now if a *-homomorphism $\sigma\colon \mathcal{S}(H_2, K_2) \to A^{\mathcal{U}}$ fulfills (2) for (H_2, K_2), then its restriction to $\mathcal{S}(H_1, K_1)$ fulfills (2) for (H_1, K_1).

Let $\{(H_i, K_i)\}_{i\in I}$ be an upward directed family with dense union in (H, K). Identify $\mathcal{S}(H_i, K_i)$ as C*-subalgebras of $\mathcal{S}(H, K)$ as in the previous paragraph. Notice that they form an upward directed family of C*-subalgebras with dense union. For each i we have a *-homomorphism $\sigma_i\colon \mathcal{S}(H_i, K_i) \to A^{\mathcal{U}}$ fulfilling (2)

(where we pick a common ultrapower for all σ_i). Pick an ultrafilter $\mathcal{V}$ in I containing the tail sets $\{i \geq i_0\}$. For $x \in \bigcup_i \mathcal{S}(H_i, K_i)$, define

$$\sigma(x) = [(\sigma_i(x))_i] \in (A^{\mathcal{U}})^{\mathcal{V}},$$

where $[(y_i)_i]$ on the right-hand side denotes the class of $(y_i)_i$ in the ultrapower. (Note that $\sigma_i(x)$ is defined for $\mathcal{V}$-almost all i.) This is a well defined and contractive *-homomorphism on $\bigcup_i \mathcal{S}(H_i, K_i)$, so it extends to $\sigma\colon \mathcal{S}(H, K) \to (A^{\mathcal{U}})^{\mathcal{V}}$. The verification that σ satisfies (2), and thus witnesses selflessness of (H, K) is left to the reader. □

In the next lemma we examine what happens when we iterate the construction of Shlyakhtenko's C*-algebra of semicirculars. We first introduce the set-up for the lemma.

Let (H_1, K_1) and (H_2, K_2) be C^*-correspondences over A with real structures. Form

$$C_1 = \mathcal{S}_A(H_1, K_1) = C^*(A, \{s_v^{(1)} : v \in K_1\}),$$

with vacuum expectation $E_1\colon C_1 \to A$. Consider the C_1-C_1 C*-correspondence induced by H_2:

$$\widetilde{H}_2 = L^2(C_1, E_1) \otimes_A H_2 \otimes_A C_1,$$

endowed with the real structure $\widetilde{K}_2$ generated by $1 \otimes K_2 \otimes 1$. Now form

$$\begin{aligned}\mathcal{S}_{C_1}(\widetilde{H}_2, \widetilde{K}_2) &= C^*(C_1, \{s^{(2)}_{1\otimes w\otimes 1} : w \in K_2\}) \\ &= C^*(A, \{s_v^{(1)}, s^{(2)}_{1\otimes w\otimes 1} : v \in K_1,\, w \in K_2\}),\end{aligned}$$

with vacuum expectation $\widetilde{E}_2\colon \mathcal{S}_{C_1}(\widetilde{H}_2, \widetilde{K}_2) \to C_1$. Endow $\mathcal{S}_{C_1}(\widetilde{H}_2, \widetilde{K}_2)$ with the expectation $E_1\widetilde{E}_2$ onto A.

Lemma 3.7. *We have canonical isomorphisms*

$$\mathcal{S}_{C_1}(\widetilde{H}_2, \widetilde{K}_2) \cong \mathcal{S}_A(H_1 \oplus H_2, K_1 \oplus K_2) \cong \mathcal{S}_A(H_1, K_1) *_A \mathcal{S}_A(H_2, K_2)$$

which intertwine the expectations onto A. Moreover, under the first isomorphism the semicircular generators are mapped as follows:

$$\begin{aligned} s_v^{(1)} &\mapsto s_{v\oplus 0}, \qquad v \in K_1, \\ s^{(2)}_{1\otimes w\otimes 1} &\mapsto s_{0\oplus w}, \qquad w \in K_2. \end{aligned}$$

Proof. Set $D = \mathcal{S}_{C_1}(\widetilde{H}_2, \widetilde{K}_2)$. Let us show that

$$L^2(D, E_1\widetilde{E}_2) \cong \mathcal{F}_A(H_1 \oplus H_2)$$

as C*-correspondences over A, via an explicit unitary.

Recall that, as remarked in (3),

$$L^2(C_1, E_1) \cong \mathcal{F}_A(H_1), \quad L^2(D, \widetilde{E}_2) \cong \mathcal{F}_{C_1}(\widetilde{H}_2),$$

where the first isomorphism is an isomorphism of C_1-A C*-correspondences, and the second one of D-C_1 C*-correspondences. From the first isomorphism we get that

$$\widetilde{H}_2 \cong \mathcal{F}_A(H_1) \otimes_A H_2 \otimes_A C_1$$

as C_1-C_1 C*-correspondences. From the second one we get that

$$\begin{aligned} L^2(D, E_1\widetilde{E}_2) &\cong \mathcal{F}_{C_1}(\widetilde{H}_2) \otimes_{C_1} L^2(C_1, E_1) \\ &\cong \mathcal{F}_{C_1}(\widetilde{H}_2) \otimes_{C_1} \mathcal{F}_A(H_1) \end{aligned}$$

as D-A C*-correspondences.

Since $\widetilde{H}_2 = \mathcal{F}_A(H_1) \otimes_A H_2 \otimes_A C_1$, the n-th tensor power $\widetilde{H}_2^{\otimes_{C_1} n}$ has the form

$$\mathcal{F}_A(H_1) \otimes_A H_2 \otimes_A \mathcal{F}_A(H_1) \otimes_A H_2 \otimes_A \cdots \otimes_A \mathcal{F}_A(H_1) \otimes_A H_2 \otimes_A C_1,$$

where $\mathcal{F}_A(H_1)$ and H_2 appear n times. Using now that $L^2(D, E_1\widetilde{E}_2) \cong \mathcal{F}_{C_1}(\widetilde{H}_2) \otimes \mathcal{F}_A(H_1)$, we get that

$$L^2(D, E_1\widetilde{E}_2) \cong \bigoplus_{n=0}^{\infty} \mathcal{F}_A(H_1) \otimes_A H_2 \otimes_A \mathcal{F}_A(H_1) \otimes_A H_2 \cdots \otimes_A \mathcal{F}_A(H_1),$$

where $\mathcal{F}_A(H_1)$ appears $n+1$ times and H_2 appears n times in the n-th summand on the right-hand side. By expanding each tensor factor $\mathcal{F}_A(H_1)$ into a direct sum of tensor powers of H_1, we can identify the infinite direct sum on the right-hand side with $\mathcal{F}_A(H_1 \oplus H_2)$. The unitary

$$U \colon L^2(D, E_1\widetilde{E}_2) \to \mathcal{F}_A(H_1 \oplus H_2)$$

is defined on elementary tensors by interpreting each H_1-word as a block of H_1-letters and each H_2-factor as an H_2-letter, thereby obtaining a tensor word in $\mathcal{F}_A(H_1 \oplus H_2)$.

Under U, the copy of C_1 in D acts as the canonical copy of $\mathcal{S}_A(H_1, K_1)$ in $\mathcal{S}_A(H_1 \oplus H_2, K_1 \oplus K_2)$. Thus, for $v \in K_1$,

$$U s_v^{(1)} U^* = s_{v \oplus 0}.$$

Also, for $w \in K_2$, the creation operator associated to $1 \otimes w \otimes 1 \in \widetilde{H}_2$ is carried by U to the creation operator associated to $0 \oplus w \in H_1 \oplus H_2$, and likewise for the adjoints of these operators. Hence

$$U s_{1 \otimes w \otimes 1}^{(2)} U^* = s_{0 \oplus w}.$$

Therefore

$$UDU^* = \mathcal{S}_A(H_1 \oplus H_2, K_1 \oplus K_2),$$

and $E_1\widetilde{E}_2$ corresponds to the vacuum expectation. This gives the first isomorphism.

For the second isomorphism, we recall that by a result of Speicher, there is an expectation preserving isomorphism

$$\mathcal{T}_A(H_1 \oplus H_2) \cong \mathcal{T}_A(H_1) *_A \mathcal{T}_A(H_2).$$

[4, Theorem 2.4]. On left creation operators, this isomorphism makes the assignments:

$$\begin{aligned} \ell_{v \oplus 0} &\to \ell_v \in \mathcal{T}_A(H_1), \qquad v \in H_1, \\ \ell_{0 \oplus w} &\to \ell_w \in \mathcal{T}_A(H_2), \qquad w \in H_2. \end{aligned}$$

Restricting to the C*-subalgebras generated by semicircular elements, we obtain the required isomorphism. This completes the proof. See [33, Proposition 2.18] for the von Neumann algebras version. □

Theorem 3.8. *If (H, K) is selfless, then so is $(\ell^2(H), \ell^2(K))$.*

Proof. By Theorem 3.6, it suffices to show that if (H, K) is selfless, then so is $(H \oplus H, K \oplus K)$. Let us prove this.

As in the paragraphs above, we set $C = \mathcal{S}(H, K)$, let $E\colon C \to A$ be the vacuum expectation, and denote by (H_C, K_C) the C*-correspondence over C induced by (H, K). Since (H, K) is selfless, we have, by Lemma 3.5, a covariant map $\Psi\colon H_C \to H^{\mathcal{U}}$ with underlying *-homomorphism $\sigma\colon C \to A^{\mathcal{U}}$ satisfying that $\Psi\Phi = \Delta_H$ and $\sigma|_A = \Delta_A$. By Toeplitz exactness (Lemma 2.2), there is a *-homomorphism $\tilde{\sigma}\colon \mathcal{T}(H_C) \to \mathcal{T}(H)^{\mathcal{U}}$ satisfying that

- $\tilde{\sigma}|_C = \sigma$,
- $\tilde{\sigma}$ intertwines expectations, i.e.,
$$E^{\mathcal{U}} \circ \tilde{\sigma} = \sigma \circ \tilde{E},$$
where $\tilde{E}\colon \mathcal{T}(H_C) \to C$ is the vacuum expectation.
- if $\tilde{v} = \Phi(v) = 1 \otimes v \otimes 1$, then $\tilde{\sigma}(\ell_{\tilde{v}})$ is the image of the constant sequence $(\ell_v)_i$ in $\mathcal{T}(H)^{\mathcal{U}}$, and consequently $\tilde{\sigma}(s_{\tilde{v}}) = \Delta_C(s_v)$.

If $v \in K$ and $c \in C$, and we set $\tilde{v} = c^* \otimes v \otimes c$, then

$$\tilde{\sigma}(s_{\tilde{v}}) = \sigma(c)^* \Delta_C(s_v) \sigma(c) \in C^{\mathcal{U}}.$$

Thus, the restriction of $\tilde{\sigma}$ to

$$D = C^*(C, \{s_{\tilde{v}} : \tilde{v} \in K_C\}) \cong \mathcal{S}(H \oplus H, K \oplus K)$$

ranges in $C^{\mathcal{U}}$. Composing with $\sigma^{\mathcal{U}}$, we get $\sigma^{\mathcal{U}}\tilde{\sigma}|_D\colon D \to (A^{\mathcal{U}})^{\mathcal{U}}$ that restricts to the diagonal embedding on A.

Let us check that $\sigma^{\mathcal{U}}\tilde{\sigma}|_D$ intertwines the coefficient maps of $H \oplus H$ as in (2). Let $v, w \in H$. By selflessness of (H, K),

$$\kappa^{\mathcal{U}}_{v,w} \circ \sigma = \Delta_A \circ \kappa_{v,w} \circ E.$$

Applying this identity in the ultrapower and using

$$E^{\mathcal{U}} \circ \tilde{\sigma} = \sigma \circ \tilde{E},$$

we get

$$\begin{aligned}(\kappa^{\mathcal{U}}_{v,w})^{\mathcal{U}} \circ (\sigma^{\mathcal{U}}\tilde{\sigma}) &= \Delta^{\mathcal{U}}_A \circ \kappa^{\mathcal{U}}_{v,w} \circ E^{\mathcal{U}} \circ \tilde{\sigma} \\ &= \Delta^{\mathcal{U}}_A \circ \kappa^{\mathcal{U}}_{v,w} \circ \sigma \circ \tilde{E} \\ &= \Delta^{\mathcal{U}}_A \circ \Delta_A \circ \kappa_{v,w} \circ E \circ \tilde{E}.\end{aligned}$$

Identifying the iterated diagonal embedding with the diagonal embedding into $(A^{\mathcal{U}})^{\mathcal{U}}$, this gives

$$(\kappa^{\mathcal{U}}_{v,w})^{\mathcal{U}} \circ (\sigma^{\mathcal{U}}\tilde{\sigma}) = \Delta_A \circ \kappa_{v,w} \circ E \circ \tilde{E}.$$

Under the identification

$$D \cong \mathcal{S}(H \oplus H, K \oplus K),$$

from Lemma 3.7, the expectation onto A is $E \circ \tilde{E}$. For the subspaces $H \oplus 0$ and $0 \oplus H$, the coefficient maps are $\kappa_{v,w}$, with $v, w \in H$, while the cross coefficient maps are zero. Hence Lemma 3.4 shows that $\sigma^{\mathcal{U}}\tilde{\sigma}|_D$ satisfies (2), witnessing selflessness of $(H \oplus H, K \oplus K)$. □

Next we formulate a simpler, but much more restrictive, variation on selflessness.

Definition 3.9. *Let H be a C*-correspondence over A. We call H Toeplitz selfless if the embedding*

$$(A; \{\kappa_{v,w} : v, w \in H\}) \to (\mathcal{T}(H); \{\kappa_{v,w} \circ E : v, w \in H\}) \tag{4}$$

is positively existential.

Note that if H is Toeplitz selfless, then (H, K) is selfless with $K = H$, but the latter property is in principle weaker. Lemmas 3.3 and 3.4 remain valid in this case. Similarly to Lemma 3.5, H is Toeplitz selfless if and only if the covariant embedding

$$H \to L^2(\mathcal{T}(H), E) \otimes_A H \otimes_A \mathcal{T}(H), \qquad v \mapsto 1 \otimes v \otimes 1,$$

is a positively existential embedding of C*-correspondences. The version of Theorem 3.6 for Toeplitz selflessness is already implicitly proved in that same theorem. We also have the following:

Lemma 3.10. *If H is Toeplitz selfless, then the embedding (4) is completely positively existential (Definition 2.4). Thus, the notions of Toeplitz selfless and completely Toeplitz selfless agree.*

Proof. Let H be Toeplitz selfless, and $\sigma\colon \mathcal{T}(H) \to A^{\mathcal{U}}$ be such that $\sigma|_A = \Delta_A$.

Let C be a unital C*-algebra. Let $\theta\colon A \to A \otimes C$ denote the first factor embedding $a \mapsto a \otimes 1$ into the minimal tensor product $A \otimes C$. Let $H \boxtimes C$ be the exterior tensor product of H with the identity C*-correspondence C, over $A \otimes C$ (see [23, Chapter 4]).

We obtain a representation of $H \boxtimes C$ in $(A \otimes C)^{\mathcal{U}}$, with underlying *-homomorphism the diagonal embedding $\Delta_{A\otimes C}$, by mapping $v \otimes 1$ to $\sigma(\ell_v) \in A^{\mathcal{U}}$ followed by the embedding $\theta^{\mathcal{U}}\colon A^{\mathcal{U}} \to (A \otimes C)^{\mathcal{U}}$:

$$H \boxtimes C \ni v \otimes 1 \mapsto \theta^{\mathcal{U}}\sigma(\ell_v) \in (A \otimes C)^{\mathcal{U}}. \tag{5}$$

Indeed, using that

$$\Delta_A(\langle v, aw\rangle) = \sigma(\ell_v)^* \Delta_A(a)\sigma(\ell_w)$$

for all $a \in A$, and that the range of $\theta^{\mathcal{U}}$ commutes with $\Delta_{A\otimes C}(1 \otimes C)$ in $(A \otimes C)^{\mathcal{U}}$, we get

$$\begin{aligned}\Delta_{A\otimes C}(\langle (v \otimes 1), (a \otimes c)(w \otimes 1)\rangle) &= \Delta_{A\otimes C}(\langle v, aw\rangle \otimes c) \\ &= (\theta^{\mathcal{U}}\sigma(\ell_v))^* \Delta_{A\otimes C}(a \otimes c)(\theta^{\mathcal{U}}\sigma(\ell_w)).\end{aligned}$$

Since the set $\{v \otimes 1 : v \in H\}$ generates $H \boxtimes C$ as an $A \otimes C$-$A \otimes C$ correspondence, the assignment (5) extends to a representation of $H \boxtimes C$ in $(A \otimes C)^{\mathcal{U}}$.

From the representation of $H \boxtimes C$ in $(A \otimes C)^{\mathcal{U}}$ we obtain, by the universal property of the Toeplitz-Pimsner C*-algebra, a *-homomorphism

$$\tilde{\sigma}\colon \mathcal{T}(H \boxtimes C) \to (A \otimes C)^{\mathcal{U}}$$

such that $\tilde{\sigma}(\ell_{v\otimes 1}) = \theta^{\mathcal{U}}\sigma(\ell_v)$ and $\tilde{\sigma}|_{A\otimes C} = \Delta_{A\otimes C}$. On the other hand

$$\mathcal{T}(H \boxtimes C) \cong \mathcal{T}(H) \otimes C$$

canonically via the assignment $\ell_{v\otimes c} \mapsto \ell_v \otimes c$ ([5, Lemma 4.6.24]). We thus obtain $\tilde{\sigma}\colon \mathcal{T}(H) \otimes C \to (A \otimes C)^{\mathcal{U}}$ such that $\tilde{\sigma}(x \otimes 1) = \theta^{\mathcal{U}}\sigma(x)$ and $\tilde{\sigma}|_{A\otimes C} = \Delta_{A\otimes C}$. This shows that the embedding $A \to \mathcal{T}(H)$ is completely positively existential, in the sense of Definition 2.4 and of [25, Section 6]. □

The proof of the following theorem can be adapted from the proof of Theorem 3.8, and it is simpler:

Theorem 3.11. *If H is Toeplitz selfless, then so is $\ell^2(H)$.*

4. Weak selflessness and weak containment

Definition 4.1. *We call (H, K) (completely) weakly selfless if the embedding*

$$A \to \mathcal{S}(\ell^2(H), \ell^2(K))$$

is a (completely) positively existential embedding of C-algebras. In other words, if there exists a *-homomorphism $\sigma\colon \mathcal{S}(\ell^2(H), \ell^2(K)) \to A^{\mathcal{U}}$ such that $\sigma|_A = \Delta_A$.*

We call H weakly Toeplitz selfless if the embedding

$$A \to \mathcal{T}(\ell^2(H))$$

is a positively existential embedding of C-algebras.*

By Theorems 3.8 and 3.11, (Toeplitz) selflessness implies weak (Toeplitz) selflessness.

Definition 4.2. *Let (H_1, K_1) and (H_2, K_2) be C*-correspondences over A with real structures. We say that (H_1, K_1) is weakly contained in (H_2, K_2), denoted by $(H_1, K_1) \precsim (H_2, K_2)$, if for every pair of finite sets $F \subset A$, $G \subset K_1$, and $\epsilon > 0$, there exists $n \in \mathbb{N}$ and an assignment*

$$G \ni v \mapsto (\xi_1(v), \cdots, \xi_n(v)) \in K_2^n$$

such that

$$\left\| \langle v, aw \rangle - \sum_{k=1}^{n} \langle \xi_k(v), a\xi_k(w) \rangle \right\| < \epsilon$$

for all $v, w \in G$ and $a \in F$. If $K_1 = H_1$ and $K_2 = H_2$ then we say that H_1 is weakly contained in H_2, and write $H_1 \precsim H_2$.

It is not hard to show that weak containment can be reformulated as follows:

Lemma 4.3. *We have $(H_1, K_1) \precsim (H_2, K_2)$ if and only if for some ultrafilter $\mathcal{U}$ there is a covariant embedding*

$$(\Phi, \Delta_A)\colon (\ell^2(H_1), \ell^2(K_1), A) \to ((\ell^2(H_2))^{\mathcal{U}}, (\ell^2(K_2))^{\mathcal{U}}, A^{\mathcal{U}}).$$

In the following theorem std denotes the real structure on A, viewed as the identity C*-correspondence, generated by 1. Thus, $\mathrm{std} = A_{sa}$.

Theorem 4.4. *If (H, K) is weakly selfless, then $(H, K) \precsim (A, \mathrm{std})$.*

Proof. Let $C = \mathcal{S}(\ell^2(H), \ell^2(K))$, and denote by $E\colon C \to A$ the vacuum expectation. Fix finite sets

$$\{v_1, \ldots, v_n\} \subseteq K, \qquad F \subseteq A,$$

and let $\epsilon > 0$. Set $\kappa_{ij}(a) = \langle v_i, av_j \rangle$.

For each i and each $k \in \mathbb{N}$, let $v_i^{(k)} \in \ell^2(K)$ be the copy of v_i in the k-th summand of $\ell^2(H)$, and let $s_i^{(k)} = s_{v_i^{(k)}} \in C$. After identifying $C \cong \mathcal{S}(H, K)^{*_A \infty}$ (Lemma 3.7), the elements

$$s_i^{(k)} a s_j^{(k)} - \kappa_{ij}(a)$$

are centered and belong to freely independent copies of $\mathcal{S}(H,K)$ in C. By Voiculescu's inequality [19, Proposition 7.4],

$$\Big\|\frac{1}{N}\sum_{k=1}^{N}\Big(s_i^{(k)}as_j^{(k)}-\kappa_{ij}(a)\Big)\Big\|=O\Big(N^{-1/2}\Big)\|a\|.$$

Choose N sufficiently large such that

$$\Big\|\frac{1}{N}\sum_{k=1}^{N}s_i^{(k)}as_j^{(k)}-\kappa_{ij}(a)\Big\|\leq\frac{\epsilon}{2}\|a\|$$

for all $a\in F$ and all $1\leq i,j\leq n$.

Since (H,K) is weakly selfless, there exists a *-homomorphism $\sigma\colon C\to A^{\mathcal{U}}$ whose restriction to A is the diagonal embedding. We thus have

$$\Big\|\frac{1}{N}\sum_{k=1}^{N}\sigma\Big(s_i^{(k)}\Big)a\sigma\Big(s_j^{(k)}\Big)\kappa_{ij}(a)\Big\|\leq\frac{\epsilon}{2}\|a\|$$

for all $a\in F$ and all $1\leq i,j\leq n$. Choose selfadjoint lifts of the elements $\sigma(s_i^{(k)})$ in the product algebra, and then pass to a single coordinate to obtain selfadjoint elements $h_i^{(k)}\in A$, for $i=1,\dots,n$ and $k=1,\dots,N$, such that

$$\Big\|\frac{1}{N}\sum_{k=1}^{N}h_i^{(k)}ah_j^{(k)}-\kappa_{ij}(a)\Big\|\leq\epsilon\|a\|$$

for all $a\in F$ and all $1\leq i,j\leq n$. Thus, the assignment

$$v_i\longmapsto\Big(\frac{1}{\sqrt{N}}h_i^{(1)},\dots,\frac{1}{\sqrt{N}}h_i^{(N)}\Big)$$

yields $(H,K)\precsim(A,\mathrm{std})$. □

Given a C*-correspondence H over A, let us call $\alpha\in CB(A)$ an extended coefficient of H if it can be obtained from the coefficient maps $\{\kappa_{v,w}:v,w\in H\}$ by linear combinations, operator norm limits, pre-compositions by left/right multiplication operators $x\overset{M_{a,b}}{\mapsto}axb$, and post-compositions by cb maps. That is, α is an operator norm limit of maps of the form

$$\sum_i\beta_i\circ\kappa_{v_i,w_i}\circ M_{a_i,b_i}$$

for $\beta_i\in CB(A)$, $v_i,w_i\in H$, $a_i,b_i\in A$.

Theorem 4.5. *Suppose that $(H_1,K_1)\precsim(H_2,K_2)$.*

(i) If (H_2,K_2) is weakly selfless, then so is (H_1,K_1).

(ii) If (H_2,K_2) is selfless, and moreover for each $v,w\in H_1$, $\kappa_{v,w}$ belongs to the set of extended coefficients of H_2, then (H_1,K_1) is selfless.

Proof. By Theorem 3.8 (or the definition of weakly selfless), we may replace (H_2,K_2) by $(\ell^2(H_2),\ell^2(K_2))$ and rename it as (H_2,K_2).

(i) From the covariant embedding $(H_1,K_1,A)\to(H_2^{\mathcal{U}},K_2^{\mathcal{U}},A^{\mathcal{U}})$ we get by Toeplitz exactness (Lemma 2.2) a *-homomorphism $\mathcal{T}(H_1)\to\mathcal{T}(H_2)^{\mathcal{U}}$ intertwining the expectations E_1 and $E_2^{\mathcal{U}}$, where E_1 and E_2 denote the vacuum expectations on $\mathcal{T}(H_1)$

and $\mathcal{T}(H_2)$, respectively. Restricting this *-homomorphism to the C*-algebra of semicircular elements, we get a *-homomorphism

$$\sigma_1 \colon \mathcal{S}(H_1, K_1) \to (\mathcal{S}(H_2, K_2))^{\mathcal{U}}.$$

(The analogous statement for the enveloping von Neumann of semicircular elements is [18, Theorem C].)

By weak selflessness of (H_2, K_2), we have a *-homomorphism $\sigma_2 \colon \mathcal{S}(H_2, K_2) \to A^{\mathcal{U}}$ agreeing with the diagonal inclusion on A. Then, the *-homomorphism

$$\sigma_2^{\mathcal{U}} \sigma_1 \colon \mathcal{S}(H_1, K_1) \to (A^{\mathcal{U}})^{\mathcal{U}}$$

agrees with the diagonal inclusion on A. Thus, (H_1, K_1) is weakly selfless.

(ii) Let $v, w \in H_2$. Since σ_1 intertwines the expectations E_1 and $E_2^{\mathcal{U}}$, we have that

$$\Delta_A \circ \kappa_{v,w} \circ E_1 = \kappa_{v,w}^{\mathcal{U}} \circ E_2^{\mathcal{U}} \sigma_1.$$

By selflessness of (H_2, K_2), σ_2 intertwines $\kappa_{v,w} \circ E_2$ and $\kappa_{v,w}^{\mathcal{U}}$. We deduce that $\sigma = \sigma_2^{\mathcal{U}} \sigma_1$ intertwines $\kappa_{v,w} \circ E_1$ and $(\kappa_{v,w}^{\mathcal{U}})^{\mathcal{U}}$. That is,

$$(\kappa_{v,w}^{\mathcal{U}})^{\mathcal{U}} \circ \sigma = \Delta_A \circ \kappa_{v,w} \circ E_1$$

for all $v, w \in H_2$. Since the set of cb maps $\alpha \in CB(A)$ that satisfy

$$(\alpha^{\mathcal{U}})^{\mathcal{U}} \sigma = \Delta_A \circ \alpha \circ E_1$$

is closed under linear combinations, operator norm limits, pre-compositions by left/right multiplication operators, and post-compositions (see the proof of Lemma 3.4), we deduce that the above equality holds for any extended coefficient of H_2. In particular, it holds for the coefficient maps of H_1. The desired result follows. □

We state a more general version:

Theorem 4.6. *Suppose that we have a covariant embedding $(H, K) \to \prod_{\mathcal{U}} (H_i, K_i)$ with underlying *-homomorphism Δ_A.*

(i) If (H_i, K_i) is weakly selfless for all i, then (H, K) is weakly selfless.

(ii) If (H_i, K_i) is selfless for all i and the coefficient maps of (H, K) belong to the extended coefficients set of $\mathcal{U}$-almost all (H_i, K_i). Then (H, K) is selfless.

Remark 4.7. Assuming separability of A, we could state this theorem differently by asking that $\lim_{\mathcal{U}} (H_i, K_i) = (H, K)$ in the Fell topology.

Proof. From the covariant embedding $(H, K) \to \prod_{\mathcal{U}} (H_i, K_i)$ and Toeplitz exactness (Lemma 2.2) we get a *-homomorphism

$$\sigma_1 \colon \mathcal{S}(H, K) \to \prod_{\mathcal{U}} \mathcal{S}(H_i, K_i)$$

intertwining expectations. From weak selflessness of (H_i, K_i) for all i, we have for each i a *-homomorphism $\sigma_{i,2} \colon \mathcal{S}(H_i, K_i) \to A^{\mathcal{V}}$ (we pick a single ultrapower for all i). From these embeddings we get $[(\sigma_{i,2})_i] \colon \prod_{\mathcal{U}} \mathcal{S}(H_i, K_i) \to (A^{\mathcal{V}})^{\mathcal{U}}$. (We use $[\cdot]$ to denote the quotient map onto the ultrapower.) Set $\tilde{\sigma} = [(\sigma_{i,2})_i] \circ \sigma_1$. If $\alpha \in CB(A)$ is an extended coefficient map for $\mathcal{U}$-almost all H_i, then

$$(\alpha^{\mathcal{V}})^{\mathcal{U}} \circ \tilde{\sigma} = \Delta_A \circ \alpha \circ E,$$

where $E \colon \mathcal{S}(H, K) \to A$ is the vacuum expectation. By assumption, this holds for all the coefficient maps of H. Thus, (H, K) is selfless. □

We have a version of the previous theorem for Toeplitz selflessness. The proof is identical to that of Theorem 4.5, using the Toeplitz part of Lemma 2.2 and replacing semicircular algebras by Toeplitz–Pimsner algebras.

Theorem 4.8. *Suppose that $H_1 \precsim H_2$.*

(i) If H_2 is weakly Toeplitz selfless, then so is H_1.

(ii) If H_2 is Toeplitz selfless, and moreover for each $v, w \in H_1$, $\kappa_{v,w}$ belongs to the extended coefficients set of H_2, then H_1 is Toeplitz selfless.

5. Selfless cp maps on C^*-algebras

Let A be a unital C*-algebra and $\phi\colon A \to A$ a cp map. Denote by $A \otimes_\phi A$ the C*-correspondence over A obtained from ϕ via the KSGNS construction [23]. We note that $A \otimes_\phi A$ is singly generated by $\xi_\phi = 1 \otimes 1$ as closed bimodule.

We endow $A \otimes_\phi A$ with the real structure generated by the vector ξ_ϕ, which we call the *standard real structure*, and denote by std.

Definition 5.1. *We call a cp map $\phi\colon A \to A$ (weakly) selfless if $(A \otimes_\phi A, \mathrm{std})$ is (weakly) selfless. We call ϕ (weakly) Toeplitz selfless if $A \otimes_\phi A$ is (weakly) Toeplitz selfless.*

Let $\ell_\phi \in \mathcal{T}(A \otimes_\phi A)$ be the left creation operator associated to ξ_ϕ, and set $s_\phi = \ell_\phi + \ell_\phi^*$. Then $C^*(A, \ell_\phi) = \mathcal{T}(A \otimes_\phi A)$ and $C^*(A, s_\phi) = \mathcal{S}(A \otimes_\phi A, \mathrm{std})$. Let $E\colon C^*(A, \ell_\phi) \to A$ be the vacuum expectation. The coefficient map $\kappa_{\xi_\phi, \xi_\phi}$ associated to ξ_ϕ is precisely ϕ:

$$\kappa_{\xi_\phi,\xi_\phi}(a) = \langle 1 \otimes 1,\ a(1 \otimes 1)\rangle = \phi(a), \qquad a \in A.$$

Since $A \otimes_\phi A$ is singly generated by ξ_ϕ, and $\kappa_{\xi_\phi,\xi_\phi} = \phi$, we deduce from Lemma 3.4 that ϕ is selfless if and only if the embedding

$$(A; \phi) \hookrightarrow (C^*(A, s_\phi); \phi \circ E).$$

is positively existential. Similarly, ϕ is Toeplitz selfless if and only if

$$(A; \phi) \hookrightarrow (C^*(A, \ell_\phi); \phi \circ E).$$

is positively existential.

Definition 5.2. *Let $\phi \in CP(A)$. We call ϕ approximately inner if it is a point-norm limit of cp maps of the form*

$$x \mapsto \sum_{i=1}^n v_i^* x v_i,$$

where $v_1, \ldots, v_n \in A$. We call ϕ approximately selfadjoint-inner (a.s.a. inner) if it is a point-norm limit of cp maps of the form

$$x \mapsto \sum_{i=1}^n h_i x h_i,$$

where $h_1, \ldots, h_n \in A_{sa}$.

If ϕ is weakly selfless, then $(A \otimes_\phi A, \mathrm{std}) \precsim (A, \mathrm{std})$, by Theorem 4.4. This readily implies that ϕ is a.s.a. inner. In fact, a.s.a. innerness of ϕ holds in a uniform way, in the sense of the following theorem:

Theorem 5.3. *If $\phi \in CP(A)$ is weakly selfless, then it is a.s.a. inner. Moreover, for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that, for every finite set $F \subseteq A$, there exist selfadjoint elements $h_1, \dots, h_N \in A$ satisfying*

$$\left\| \phi(a) - \sum_{i=1}^{N} h_i a h_i \right\| \leq \epsilon \|a\|$$

for all $a \in F$. Here N depends only on ϵ, and not on F.

Proof. This follows from the proof of Theorem 4.4 applied to the singleton set $\{v\} = \{\xi_\phi\} \subset K$. The number N chosen in the application of Voiculescu's inequality depends on ϵ, but not on the elements of the finite set F. □

Given a cp map $\phi \in CP(A)$, let us call $\psi \in CP(A)$ an extended coefficient of ϕ if it is an operator norm limit of cb maps of the form

$$x \mapsto \sum_{i=1}^{n} (\beta_i \circ \phi)(a_i x b_i),$$

where $a_i, b_i \in A$ and $\beta_i \in CB(A)$. Note that in particular if $\psi = \beta \circ \phi$ for some β, then ψ is an extended coefficient of ϕ.

Theorem 5.4. *Let A be a unital C^*-algebra.*

(i) If $\phi \in CP(A)$ is weakly selfless, then so is

$$\psi : x \mapsto \sum_{k=1}^{n} \sum_{i,j=1}^{m_k} a_{k,i} \phi(a_{k,i}^* x a_{k,j}) a_{k,j}^*,$$

where $a_{k,i} \in A$. If ϕ is selfless, then the displayed map is selfless as well.

(ii) If $(\phi_i)_i \in CP(A)$ is a net of weakly selfless cp maps, and $\phi_i \to \phi$ in point-norm, then ϕ is weakly selfless. If, in addition, each ϕ_i is selfless and ϕ is an extended coefficient of ϕ_i for all i, then ϕ is selfless.

(iii) If $\phi, \psi \in CP(A)$ satisfy that ϕ is weakly selfless and ψ is a.s.a. inner, then

$$\theta := \frac{1}{2}(\phi\psi + \psi\phi),$$

is weakly selfless. If in addition ϕ is selfless, and θ is an extended coefficient of ϕ, then θ is selfless. In particular, this holds if ψ and ϕ commute.

(iv) If $\phi \in CP(A_1)$ is weakly selfless, $\psi \in CP(A_2)$ is a.s.a. inner, and A_2 is exact, then $\phi \otimes \psi$ is weakly selfless. If, moreover, ϕ is selfless, then $\phi \otimes \psi$ is selfless. (The tensor product is the minimal one.)

(v) If ϕ is a selfless unital cp map and $p \in A$ is a projection fixed by ϕ, then $\phi|_{pAp} \colon pAp \to pAp$ is selfless.

Proof. (i) For $j = 1, \dots, n$, define $v_j \in A \otimes_\phi A$ by

$$v_k = \sum_{i=1}^{m_k} a_{k,i} \xi_\phi a_{k,i}^*.$$

Now, using Lemma 2.1, embed $(A \otimes_\psi A, \mathrm{std})$ into $(A \otimes_\phi A, \mathrm{std})^{\oplus n}$ by mapping

$$\xi_\psi \mapsto (v_1, \dots, v_n).$$

Apply Theorem 4.5.

(ii) Choose an ultrafilter $\mathcal{U}$ on the index set and containing the tail sets $\{i : i \geq i_0\}$. From the point-norm convergence $\phi_i \to \phi$, we get a covariant embedding

$$(A \otimes_\phi A, \mathrm{std}) \to \prod_{\mathcal{U}} (A \otimes_{\phi_i} A, \mathrm{std})$$
$$\xi_\phi \mapsto [(\xi_{\phi_i})_i],$$

where the underlying *-homomorphism is $\Delta_A \colon A \to A^{\mathcal{U}}$. We now apply Theorem 4.6.

(iii) Note first that for selfadjoint $a \in A$, the vectors

$$p_a = \frac{1}{2}(\xi_\phi a + a\xi_\phi), \qquad q_a = \frac{1}{2i}(\xi_\phi a - a\xi_\phi).$$

belongs to std (are real) and satisfy that

$$\kappa_{p_a,p_a}(x) + \kappa_{q_a,q_a}(x) = \frac{1}{2}(a\phi(x)a + \phi(axa)).$$

Suppose that ψ is selfadjoint inner, i.e.,

$$\psi(x) = \sum_{j=1}^{n} a_j x a_j,$$

for some $a_1, \ldots, a_n$ selfadjoint. Define $p_j, q_j \in A \otimes_\phi A$ by

$$p_j = \frac{1}{2}(\xi_\phi a_j + a_j \xi_\phi), \qquad q_j = \frac{1}{2i}(\xi_\phi a_j - a_j \xi_\phi).$$

Then $(A \otimes_\theta A, \mathrm{std})$ embeds into $(A \otimes_\phi A, \mathrm{std})^{\oplus 2n}$ via mapping

$$\xi_\theta \mapsto (p_1, q_1, \ldots, p_n, q_n).$$

(Thus, θ has the form dealt with in (i)). If ψ is a.s.a. inner, we deduce that $A \otimes_\theta A$ is weakly contained in $A \otimes_\phi A$. Apply Theorem 4.5.

(iv) We first show that $\phi \otimes \mathrm{id}_{A_2}$ is weakly selfless whenever ϕ is weakly selfless, and selfless whenever ϕ is selfless. We will denote the C*-correspondence obtained from a cp map ϕ by H_ϕ, rather than $A \otimes_\phi A$ as we have done before, to keep the notation compact.

Let $C = \mathcal{S}(\ell^2(H_\phi), \ell^2(\mathrm{std}))$, and let $E \colon C \to A_1$ be the vacuum expectation. Since ϕ is weakly selfless, there exist an ultrafilter $\mathcal{U}$ and a *-homomorphism $\sigma \colon C \to A_1^{\mathcal{U}}$ such that $\sigma|_{A_1} = \Delta_{A_1}$. If ϕ is selfless, then, by the ℓ^2-stability of selflessness, σ may moreover be chosen so that

$$\phi^{\mathcal{U}} \circ \sigma = \Delta_{A_1} \circ \phi \circ E.$$

Tensoring σ with id_{A_2} gives a *-homomorphism

$$\sigma \otimes \mathrm{id}_{A_2} \colon C \otimes A_2 \to A_1^{\mathcal{U}} \otimes A_2.$$

Since A_2 is exact, there is a canonical embedding $A_1^{\mathcal{U}} \otimes A_2 \to (A_1 \otimes A_2)^{\mathcal{U}}$. We therefore obtain a *-homomorphism $\widetilde{\sigma} \colon C \otimes A_2 \to (A_1 \otimes A_2)^{\mathcal{U}}$.

There is a natural isomorphism

$$\mathcal{S}(\ell^2(H_\phi), \ell^2(\mathrm{std})) \otimes A_2 \cong \mathcal{S}(\ell^2(H_{\phi \otimes \mathrm{id}_{A_2}}), \ell^2(\mathrm{std})),$$

under which the expectation $E \otimes \mathrm{id}_{A_2}$ on the left-hand side corresponds to the vacuum expectation on the right-hand side. (This is proven by first obtaining the corresponding isomorphism for Toeplitz-Pimsner C*-algebras using their universal property, and then observing that it restricts to the isomorphism displayed above.)

Since $\widetilde{\sigma}$ restricts to the diagonal embedding on $A_1 \otimes A_2$, this proves that $\phi \otimes \mathrm{id}_{A_2}$ is weakly selfless. In the selfless case, the canonical embedding $A_1^{\mathcal{U}} \otimes A_2 \to (A_1 \otimes A_2)^{\mathcal{U}}$ intertwines $\phi^{\mathcal{U}} \otimes \mathrm{id}_{A_2}$ with $(\phi \otimes \mathrm{id}_{A_2})^{\mathcal{U}}$. Consequently,

$$(\phi \otimes \mathrm{id}_{A_2})^{\mathcal{U}} \circ \widetilde{\sigma} = \Delta_{A_1 \otimes A_2} \circ (\phi \otimes \mathrm{id}_{A_2}) \circ (E \otimes \mathrm{id}_{A_2}).$$

Thus $\phi \otimes \mathrm{id}_{A_2}$ is selfless.

Since ψ is a.s.a. inner, so is $\mathrm{id}_{A_1} \otimes \psi$. Moreover, $\phi \otimes \mathrm{id}_{A_2}$ and $\mathrm{id}_{A_1} \otimes \psi$ commute, and

$$\phi \otimes \psi = (\phi \otimes \mathrm{id}_{A_2}) \circ (\mathrm{id}_{A_1} \otimes \psi).$$

The result follows from part (iii).

(v) We have

$$C^*(pAp, s_{\phi|_{pAp}}) \subset pC^*(A, s_\phi)p \subset (pAp)^{\mathcal{U}}.$$

□

Remark 5.5. The previous theorem holds with "completely selfless" in place of "selfless", and in this case we may remove the assumption of exactness of A_2 in (iv).

Corollary 5.6. *If $\phi \in CP(A)$ is (weakly) selfless, then so is ϕ^k for all $k = 1, 2, \ldots$. Moreover, if the limit*

$$\frac{1}{N} \sum_{k=1}^{N} \phi^k \to P$$

exists in point-norm, then P is (weakly) selfless.

Proof. Let $k \geq 1$. Suppose, inductively, that ϕ^k is (weakly) selfless (the case $k = 1$ is our hypothesis). Since ϕ is a.s.a. inner, and ϕ and ϕ^k commute, $\phi^{k+1} = \phi^k \circ \phi$ is (weakly) selfless by Theorem 5.4 (iii).

Suppose that the displayed point norm limit exists. Then P is a.s.a. inner, as it is a point-norm limit of a.s.a. inner maps, and $\phi P = P\phi = P$. By Theorem 5.4 (iii), P is (weakly) selfless. □

Corollary 5.7. *Let G be a countable discrete group. Let $v \in \ell^2(G; \mathbb{R})$, and let $\phi_v \colon C_r^*(G) \to C_r^*(G)$ be the Fourier multiplier map associated to $g \mapsto \langle v, \lambda_g v \rangle$, i.e.,*

$$\phi_v(\lambda_g) = \langle v, \lambda_g v \rangle \lambda_g,$$

for all $g \in G$. If $(C_r^*(G), \tau)$ is (completely) selfless (where τ denotes the canonical trace on $C_r^*(G)$), then ϕ_v is (completely) selfless.

Proof. The assumption of (complete) selflessness of $(C_r^*(G), \tau)$ means that the trace τ is (completely) selfless when viewed as a cp map on $C_r^*(G)$ (see Theorem 7.3). We can embed $(C_r^*(G) \otimes_{\phi_v} C_r^*(G), \mathrm{std})$ in $(C_r^*(G) \otimes_\tau C_r^*(G), \mathrm{std})$ by the assignment

$$\xi_{\phi_v} \mapsto \sum_{g \in G} v_g (\lambda_g \otimes \lambda_g^*).$$

(Indeed, a quick calculation shows that the vector on the right-hand side gives rise to ϕ_v as a coefficient map.) The result now follows from Theorem 3.6. □

The following is a cp-map analogue of Ozawa's selflessness criterion [25, Lemma 12 and Theorem 13] for the case of a state.

Proposition 5.8. *Let $\phi \in CP(A)$. Suppose that there exist a unital inclusion $A \subseteq C$, a cp map $\psi \in CP(C)$ extending ϕ to C, and $v \in C^{\mathcal{U}}$ (for some ultrafilter) such that*

(1) $v^*av = \phi(a)$ for all $a \in A$,
(2) $\psi^{\mathcal{U}}(avv^*a^*) = 0$ for all $a \in A$,
(3) $v + v^* \in A^{\mathcal{U}}$.

Then ϕ is selfless.

Proof. From $v^*av = \phi(a)$ we get a representation of $A \otimes_\phi A$ in $C^{\mathcal{U}}$ (a covariant embedding into the identity correspondence of $C^{\mathcal{U}}$) such that $\xi_\phi \mapsto v$, and with underlying *-homomorphism the embedding of A in $C^{\mathcal{U}}$ (Lemma 2.1). Thus, by the universal property of the Toeplitz-Pimsner C*-algebra, there exists a *-homomorphism $\sigma\colon \mathcal{T}(A \otimes_\phi A) \to C^{\mathcal{U}}$ such that $\sigma|_A$ is the diagonal embedding of A in $C^{\mathcal{U}}$ and $\sigma(\ell_\phi) = v$. Since $v + v^* \in A^{\mathcal{U}}$, the restriction of σ to $C^*(A, s_\phi)$ ranges in $A^{\mathcal{U}}$.

Consider $\tilde{\phi} = \psi^{\mathcal{U}}\sigma\colon \mathcal{T}(A \otimes_\phi A) \to C^{\mathcal{U}}$. From $\psi^{\mathcal{U}}(avv^*a^*) = 0$ for all $a \in A$ we deduce that $\tilde{\phi}(a\ell_\phi\ell_\phi^*a^*) = 0$ for all $a \in A$. By Cauchy-Schwarz, we obtain that

$$\tilde{\phi}(a\ell_\phi x) = \tilde{\phi}(x\ell_\phi^* a) = 0$$

for all $a \in A$ and $x \in \mathcal{T}(A \otimes_\phi A)$. Together with the fact that $\tilde{\phi}|_A = \Delta_A \circ \phi|_A$, this implies that $\tilde{\phi} = \Delta_A \circ \phi \circ E$, i.e., $\psi^{\mathcal{U}} \circ \sigma = \Delta_A \circ \phi \circ E$, where $E\colon \mathcal{T}(A \otimes_\phi A) \to A$ denotes the vacuum expectation. Since ψ extends ϕ and $\sigma|_{C^*(A,s_\phi)}$ ranges in $A^{\mathcal{U}}$, we get $\phi^{\mathcal{U}} \circ \sigma|_{C^*(A,s_\phi)} = \Delta_A \circ \phi \circ E$, as desired. □

We formulate a finite sets and ϵ version of the above criterion:

Proposition 5.9. *Let $\phi \in CP(A)$. Suppose that there exist a unital inclusion $A \subseteq C$ and a cp map $\psi \in CP(C)$ extending ϕ to C such that for every finite set $F \subset A$ and $\epsilon > 0$, there exists $v \in C$ such that*

(1) $\|v^*av - \phi(a)\| < \epsilon$ for all $a \in F$,
(2) $\|\psi(avv^*a^*)\| < \epsilon$ for all $a \in F$,
(3) $\|v + v^* - s\| < \epsilon$ for some $s \in A$.

Then ϕ is selfless.

We finish this section with variants of the previous results for (weak) Toeplitz selflessness. The arguments run along similar, but simpler lines.

Proposition 5.10. *If ϕ is weakly Toeplitz selfless, then it is 1-step approximately inner. That is, ϕ is a point-norm limit of cp maps maps of the form $x \mapsto v^*xv$.*

Proof. Let ξ_1 be the first copy of ξ_ϕ in $\ell^2(A \otimes_\phi A)$, and let ℓ_1 be the corresponding creation operator in $\mathcal{T}(\ell^2(A \otimes_\phi A))$. Then $\ell_1^* a \ell_1 = \phi(a)$ for all $a \in A$. Applying $\sigma\colon \mathcal{T}(\ell^2(A \otimes_\phi A)) \to A^{\mathcal{U}}$ witnessing weak Toeplitz selflessness, we get

$$\sigma(\ell_1)^* a \sigma(\ell_1) = \phi(a)$$

for all $a \in A$. Thus, ϕ is approximately 1-step inner. □

We have a version of Theorem 5.4 for Toeplitz selflessness:

Theorem 5.11. *Let A be a unital C*-algebra.*

(i) *If $\phi \in CP(A)$ is (completely, weakly) Toeplitz selfless, then so is*

$$\psi : x \mapsto \sum_{k=1}^{n} \sum_{i,j=1}^{m_k} b_{k,i}\phi(a_{k,i}^* x a_{k,j}) b_{k,j}^*,$$

where $a_{k,i}, b_{k,i} \in A$.

(ii) *If $(\phi_i)_i \in CP(A)$ is a net of weakly Toeplitz selfless cp maps, and $\phi_i \to \phi$ in point-norm, then ϕ is weakly Toeplitz selfless. If, in addition, each ϕ_i is Toeplitz selfless and ϕ is an extended coefficient of ϕ_i for all i, then ϕ is Toeplitz selfless.*

(iii) *If $\phi, \psi \in CP(A)$ satisfy that ϕ is (weakly) Toeplitz selfless and ψ is approximately inner, then $\psi\phi$ is (weakly) Toeplitz selfless.*

(iv) *If $\phi \in CP(A_1)$ is weakly Toeplitz selfless and $\psi \in CP(A_2)$ is approximately inner, then $\phi \otimes \psi$ is weakly Toeplitz selfless. If, moreover, ϕ is Toeplitz selfless, then $\phi \otimes \psi$ is Toeplitz selfless.*

(v) *If ϕ is unital and Toeplitz selfless, and $p \in A$ is a projection fixed by ϕ, then $\phi|_{pAp} \colon pAp \to pAp$ is Toeplitz selfless.*

In the case of Toeplitz selflessness, the criterion Proposition 5.8 takes on a much simpler form:

Proposition 5.12. *A cp map $\phi \in CP(A)$ is Toeplitz selfless if and only if there exists $v \in A^{\mathcal{U}}$ such that $v^*av = \phi(a)$ and $\phi^{\mathcal{U}}(avv^*a^*) = 0$ for all $a \in A$.*

6. Some regularity properties

In this section we explore some regularity properties of A when (A, ϕ) is weakly selfless. The situation is most satisfactory in the case of Toeplitz selflessness, so we deal with this case first.

In [21] Kirchberg and Rørdam introduce the notion of strongly purely infinite C*-algebra. A C*-algebra A is strongly purely infinite if for all $a_1, a_2 \in A_+$ and $\epsilon > 0$ there exist $d_1, d_2 \in A$ such that

$$\|d_1^* a_1 d_1 - a_1\| < \epsilon, \quad \|d_2^* a_2 d_2 - a_2\| < \epsilon, \quad \|d_2^* a_2^{\frac{1}{2}} a_1^{\frac{1}{2}} d_1\| < \epsilon.$$

This is a strengthening of the notion of purely infinite C*-algebra in the non-simple case. By [21, Theorem 4.16], if A is separable and nuclear, then it is strongly purely infinite if and only if $A \otimes \mathcal{O}_\infty \cong A$.

Theorem 6.1. *If (A, ϕ) is weakly Toeplitz selfless and* $\mathrm{Ideal}(\phi(a)) = \mathrm{Ideal}(a)$ *for all $a \in A_+$, then A is strongly purely infinite.*

Proof. We start working in $C = \mathcal{T}(\ell^2(A \otimes_\phi A))$ then pass to $A^{\mathcal{U}}$ via $\sigma \colon C \to A^{\mathcal{U}}$ such that $\sigma|_A = \Delta_A$.

Let us first show that $a \precsim \phi(a)$, in the Cuntz preorder of C, for all $a \in A_+$. Let $a \in A_+$ and $\epsilon > 0$. By assumption, a belongs to the ideal generated by $\phi^2(a)$. Thus,

$$\left\| a - \sum_{i=1}^{n} x_i^* \phi^2(a) x_i \right\| < \epsilon,$$

for some $x_i \in A$. Now choose left creation operators $\ell_1, \ldots, \ell_n \in C$ corresponding to pairwise orthogonal copies $\xi_\phi^{(1)}, \ldots, \xi_\phi^{(n)}$ of the canonical generator $\xi_\phi \in A \otimes_\phi A$.

Set $x = \sum_{i=1}^n \ell_i x_i$. Then

$$x^*\phi(a)x = \sum_{i=1}^n x_i^*\ell_i^*\phi(a)\ell_i x_i = \sum_{i=1}^n x_i^*\phi^2(a)x_i,$$

where we have used that $\ell_i^* b \ell_j = 0$ if $i \neq j$ and $b \in A$. Thus, $\|a - x^*\phi(a)x\| < \epsilon$. This shows that $a \precsim \phi(a)$.

Let $a_1, a_2 \in A_+$ and $\epsilon > 0$. Choose left creation operators $\ell_1, \ell_2 \in C$ as before. Then,

$$\ell_1^* a_1 \ell_1 = \phi(a_1), \quad \ell_2^* a_2 \ell_2 = \phi(a_2), \quad \ell_2^* a_2^{\frac{1}{2}} a_1^{\frac{1}{2}} \ell_1 = 0.$$

Since $a_1 \precsim \phi(a_1)$, we can find $x_1 \in C$ such that $\|a_1 - x_1^*\phi(a_1)x_1\| < \epsilon$, and similarly we find $x_2 \in C$ such that $\|a_2 - x_2^*\phi(a_2)x_2\| < \epsilon$. Set $d_1 = \ell_1 x_1$ and $d_2 = \ell_2 x_2$. Then

$$\|d_1^* a_1 d_1 - a_1\| < \epsilon, \quad \|d_2^* a_2 d_2 - a_2\| < \epsilon, \quad d_2^* a_2^{\frac{1}{2}} a_1^{\frac{1}{2}} d_1 = 0.$$

Now we pass to $A^{\mathcal{U}}$ via σ, and then to A by projecting onto a suitable coordinate. □

The next theorem shows that if (A, ϕ) is weakly selfless, then the set $\overline{[A,A]}$ is definable in the model theory of A; see [7]. It also shows that $A^{\mathcal{U}}$ has no "phantom" traces. More concretely, for every tracial state τ on $A^{\mathcal{U}}$ and separable C*-subalgebra $D \subseteq A^{\mathcal{U}}$, there exist tracial states $(\tau_i)_i$ on A such that τ agrees with the limit trace $\lim_{\mathcal{U}} \tau_i$ on D; see [24, Theorem 8] and [2].

Theorem 6.2. *If (A, ϕ) is weakly selfless and ϕ is unital, then $\overline{[A^{\mathcal{U}}, A^{\mathcal{U}}]} = \overline{[A,A]}^{\mathcal{U}}$.*

Proof. It is enough to show that for every $\gamma > 0$ there exist $N \in \mathbb{N}$ and $M > 0$ such that, for every $a \in \overline{[A,A]}$, there exist $x_i, y_i \in A$, $i = 1, \ldots, N$, satisfying

$$\left\| a - \sum_{i=1}^N [x_i, y_i] \right\| \leq \gamma \|a\|, \qquad \|x_i\| \cdot \|y_i\| \leq M\|a\|. \tag{6}$$

Here N and M may depend on γ, but not on a. See [2, Theorem A]. We will prove the required approximation for $a \in \overline{[A,A]} \cap A_{\mathrm{sa}}$. For a general $a \in \overline{[A,A]}$, write

$$a = \mathrm{Re}(a) + i\,\mathrm{Im}(a).$$

Both terms belong to $\overline{[A,A]} \cap A_{\mathrm{sa}}$ and have norm at most $\|a\|$. Applying the selfadjoint case to each term with error factor $\gamma/2$ and adding the resulting approximations gives (6) for a.

Let $\gamma > 0$ and $a \in \overline{[A,A]} \cap A_{\mathrm{sa}}$. By the uniform form of the a.s.a. inner property of ϕ from Theorem 5.3 applied with $\epsilon = \gamma/2$ and $F = \{1, a\}$, there exist selfadjoint elements $x_1, \ldots, x_N \in A$ satisfying

$$\left\| \phi(a) - \sum_{i=1}^N x_i a x_i \right\| \leq \frac{\gamma}{2}\|a\|, \qquad \left\| 1 - \sum_{i=1}^N x_i^2 \right\| \leq \frac{\gamma}{2}.$$

Here N depends on γ but is independent of a. Since

$$a - \phi(a) = a - \sum_{i=1}^N x_i^2 a + \sum_{i=1}^N [x_i, a x_i] + \sum_{i=1}^N x_i a x_i - \phi(a),$$

we have

$$\Big\|(a-\phi(a))-\sum_{i=1}^{N}[x_i,ax_i]\Big\| \le \Big\|1-\sum_{i=1}^{N}x_i^2\Big\|\cdot\|a\|+\frac{\gamma}{2}\|a\| \le \gamma\|a\|.$$

Hence,

$$\Big\|\frac{1}{2}(a-\phi(a))-\frac{1}{2}\sum_{i=1}^{N}[x_i,ax_i]\Big\| \le \frac{\gamma}{2}\|a\|.$$

Notice that $\|x_i\|\cdot\|ax_i\| \le \|x_i\|^2\cdot\|a\| \le (1+\gamma/2)\|a\|$.

We will be done once we have shown how to approximate $\frac{1}{2}(a+\phi(a))$ by a uniformly bounded number of commutators with error at most $\frac{\gamma}{2}\|a\|$. Adding the approximations of $\frac{1}{2}(a-\phi(a))$ and $\frac{1}{2}(a+\phi(a))$ then gives (6).

The next part of the proof is inspired by the proof of [24, Theorem 6] (but we use freeness in place of commutativity in some key places). By [24, Theorem 5], there exists a universal constant $M_1>0$ such that elements of the form

$$a=\sum_{i=1}^{n}[x_i^*,x_i],$$

where $x_i\in A$ satisfy that $\sum_{i=1}^{n}\|x_i\|^2\le M_1\|a\|$, form a dense set in $\overline{[A,A]}\cap A_{sa}$. Since $\frac{1}{2}(\mathrm{id}_A+\phi)$ is contractive, density shows that it is enough to prove the required approximation of $\frac{1}{2}(a+\phi(a))$ for elements a of the form

$$a=\sum_{i=1}^{n}[x_i^*,x_i], \qquad \sum_{i=1}^{n}\|x_i\|^2\le M_1\|a\|.$$

Let us assume that a has this form.

We now work in the C*-algebra

$$C=\mathcal{S}(\ell^2(A\otimes_\phi A),\ell^2(\mathrm{std})),$$

and let $E\colon C\to A$ denote the expectation onto A. Choose a doubly indexed collection of semicircular elements $s_{ij}\in C$ associated to pairwise orthogonal copies of the canonical generator $\xi_\phi\in A\otimes_\phi A$, where $i=1,\dots,n$ and $j=1,\dots,R$ (R is chosen below). For $j=1,\dots,R$, set

$$X_j=\sum_{i=1}^{n}s_{i,j}x_i, \qquad \widetilde{X}_j=\sum_{i=1}^{n}x_is_{i,j}.$$

Observe that $E(s_{ij}x_i)=0$. Hence, $E(X_j)=0$ and similarly, $E(\widetilde{X}_j)=0$. We can estimate the norms of $X_j,\tilde{X}_j$ using Voiculescu's inequality:

$$\|X_j\|\le\sup_i\|s_{ij}x_i\|+\Big\|\sum_{i=1}^{n}x_i^*x_i\Big\|^{\frac{1}{2}}+\Big\|\sum_{i=1}^{n}x_ix_i^*\Big\|^{\frac{1}{2}}\le M_2\|a\|^{\frac{1}{2}},$$

for a universal constant M_2. Similarly, $\|\widetilde{X}_j\|\le M_2\|a\|^{\frac{1}{2}}$.

We have

$$[X_j,X_j^*]=\sum_{i=1}^{n}[s_{i,j}x_i,x_i^*s_{i,j}]+\sum_{i\neq i'}[s_{i,j}x_i,x_{i'}^*s_{i',j}].$$

Let us apply E on both sides. On the right-hand side, we use that the terms of the second sum have zero expectation and that

$$E([s_{ij}x_i,x_i^*s_{i,j}])=E(s_{ij}x_ix_i^*s_{ij})-E(x_i^*(s_{ij})^2x_i)=\phi(x_ix_i^*)-x_i^*x_i.$$

This yields

$$E([X_j, X_j^*]) = \sum_{i=1}^{n} \phi(x_i x_i^*) - x_i^* x_i.$$

A similar calculation shows that

$$E([\widetilde{X}_j, \widetilde{X}_j^*]) = \sum_{i=1}^{n} x_i x_i^* - \phi(x_i^* x_i).$$

We deduce that

$$E([X_j^*, X_j] + [\widetilde{X}_j^*, \widetilde{X}_j]) = a + \phi(a).$$

As j varies, the centered elements

$$[X_j^*, X_j] + [\widetilde{X}_j^*, \widetilde{X}_j] - (a + \phi(a))$$

belong to freely independent factors of C. Applying Voiculescu's inequality, we get

$$\begin{aligned}\Big\| \frac{1}{R} \sum_{j=1}^{R} ([X_j^*, X_j] + [\widetilde{X}_j^*, \widetilde{X}_j]) - (a + \phi(a)) \Big\| &\leq \Big(\frac{2}{R} + \frac{2}{\sqrt{R}} \Big) \sup_j \|[X_j^*, X_j] + [\widetilde{X}_j^*, \widetilde{X}_j]\| \\ &\leq \Big(\frac{2}{R} + \frac{2}{\sqrt{R}} \Big) M_3 \|a\|.\end{aligned}$$

Choosing R sufficiently large, and absorbing the coefficient $1/(2R)$ into the entries of each commutator, gives an approximation of the form (6) for $\frac{1}{2}(a + \phi(a))$, with constants independent of a.

To finish the proof, we use that by weak selflessness of ϕ there exists a *-homomorphism $\sigma \colon C \to A^{\mathcal{U}}$ that restricts to the diagonal inclusion on A. Applying σ gives the required approximation in $A^{\mathcal{U}}$. Lifting the finitely many entries of the commutators and passing to a suitable coordinate gives the corresponding approximation in A. □

7. Selfless operator-valued C^*-probability spaces

Let A be a unital C*-algebra with unital C*-subalgebra B and expectation $E \colon A \to B$. We call (A, E, B) a C*-probability space. We assume that the GNS representation of A in $L^2(A, E)$ induced by E is faithful (except for ultrapowers/ultraproducts); this ensures that the canonical maps into reduced amalgamated free products are faithful. By a morphism between C*-probability spaces we understand a unital *-homomorphism between the C*-algebras that intertwines the expectations.

We use the reduced amalgamated free product construction of C^*-probability spaces with conditional expectations; see [36, Chapter 4].

Definition 7.1. *We call (A, E, B) selfless if for some C*-probability space (C, κ), with $C \neq \mathbb{C}$ and κ a state inducing a faithful GNS, the first factor embedding*

$$A \to A *_B (B \otimes C)$$

is an existential embedding of C-probability spaces. More concretely, there exists an embedding $\sigma \colon A *_B (B \otimes C) \to A^{\mathcal{U}}$ whose restriction to A agrees with the diagonal embedding Δ_A and such that*

$$E^{\mathcal{U}} \circ \sigma = \Delta_A \circ (E *_B (\mathrm{id}_B \otimes \kappa)).$$

We call (A, E, B) Toeplitz selfless if (C, κ) may be chosen with a non-tracial κ.

In the following lemma we denote by $C^*(A, \ell_E)$ the Toeplitz-Pimsner C*-algebra of $A \otimes_E A$, i.e., $\mathcal{T}(A \otimes_E A)$, and by E_Ω its vacuum expectation. As in the previous section, we write $C^*(A, s_E)$ for $\mathcal{S}(A \otimes_E A, \mathrm{std})$. The isomorphisms of this lemma are well known. The first isomorphism appears in [5, Section 4.6, exercises] and in [4, Section 2.5], where it is attributed to Shlyakhtenko [32]. See also [25, Lemma 12]. The second isomorphism follows from the first.

Lemma 7.2. *Let (A, E, B) be a C*-probability space. Let $(\mathcal{T}, \omega)$ be the Toeplitz C*-algebra endowed with the vacuum state, and denote by $v \in \mathcal{T}$ the generating isometry. We have an isomorphism of A-valued C*-probability spaces:*

$$(C^*(A, \ell_E), E_\Omega, A) \cong (A *_B (B \otimes \mathcal{T}), \mathrm{id}_A *_B (\mathrm{id}_B \otimes \omega), A),$$

mapping the left creation operator ℓ_E to $1 \otimes v$. Restricting this isomorphism to $C^*(A, s_E)$ yields the isomorphism

$$(C^*(A, s_E), E_\Omega, A) \cong (A *_B (B \otimes C([-2, 2])), \mathrm{id}_A *_B (\mathrm{id}_B \otimes \lambda), A),$$

where the state on $C([-2, 2])$ is induced by the semicircular distribution.

Theorem 7.3. *Let (A, E, B) be a C*-probability space. The following are equivalent:*

(i) (A, E, B) is selfless.
(ii) The embedding $A \to A *_B (B \otimes C)$ is an existential embedding of B-valued C-probability spaces, where C can be any of*

$$C = C_r^*(\mathbb{F}_\infty), \qquad (C([-2, 2]))^{*\infty}.$$

(iii) The expectation $E\colon A \to B \subset A$ is a selfless cp map.

Moreover, the following are equivalent:

(v) (A, E, B) is Toeplitz selfless.
(vi) The embedding $A \to A *_B (B \otimes \mathcal{O}_\infty)$ is an existential embedding of B-valued C-probability spaces, where $\mathcal{O}_\infty$ is the Cuntz algebra endowed with the vacuum state.*
(vii) The expectation $E\colon A \to B \subset A$ is a Toeplitz selfless cp map.

Proof. (i) $\Rightarrow$ (ii): The argument runs along the same lines as the argument for the scalar case ($B = \mathbb{C}$) found in [29, Theorem 2.6]. Set $D = B \otimes C$, with expectation $\mathrm{id}_B \otimes \kappa$ onto B. Let $\sigma\colon A *_B D \to A^{\mathcal{U}}$ be a *-homomorphism such that $\sigma|_A = \Delta_A$ and intertwining the respective expectations onto B and $B^{\mathcal{U}}$. By Pisier's amalgamated free product strong convergence result [26, Theorem 7.1], or by the more general [11, Corollary 1.2], we obtain

$$\tilde{\sigma}\colon (A *_B D) *_B D \to (A *_B D)^{\mathcal{U}}$$

whose restriction to $(A *_B D)$ agrees with σ and whose restriction to the second factor D agrees with the diagonal inclusion. Post-composing with $\sigma^{\mathcal{U}}$ we obtain

$$\sigma^{\mathcal{U}}\tilde{\sigma}\colon A *_B D *_B D \to (A^{\mathcal{U}})^{\mathcal{U}}$$

that restricts to the diagonal inclusion on A and intertwines expectations. Repeated applications of this show that the embeddings $A \to A *_B D^{*_B n}$ are existential embeddings of operator valued C*-probability spaces for all n. Note that

$$D^{*_B n} \cong B \otimes (C, \kappa)^{*n}.$$

Since $C \neq \mathbb{C}$ and κ induces a faithful GNS representation, for large enough n we have that $(C_r^*(\mathbb{F}_\infty), \tau)$ embeds in $(C, \kappa)^{*n}$ ([29, Lemma 2.5]). We thus conclude that

$$A \to (A *_B (B \otimes C_r^*(\mathbb{F}_\infty)).$$

is an existential embedding of operator-valued C*-probability spaces. Since $(C([-2,2]), \lambda)^{*\infty}$ embeds in $(C_r^*(\mathbb{F}_\infty), \tau)$, we get an existential embedding in this case too.

(ii) $\Leftrightarrow$ (iii): By Lemma 7.2, $C^*(A, s_E) \cong A *_B (B \otimes C([-2,2]))$.

(iii) $\Rightarrow$ (i) is obvious.

Let us now deal with the equivalence of the last three assertions.

(vi) $\Leftrightarrow$ (vii) follows from Lemma 7.2.

(vi) $\Rightarrow$ (v) is immediate.

To prove that (v) implies (vi), recall that, as argued above, the embedding

$$A \to A *_B (B \otimes (C, \kappa)^{*n}))$$

is existential. For large enough n, $(C', \kappa') = C_r^*(\mathbb{F}_\infty) * (C, \kappa)$ embeds in $(C, \kappa)^{*n}$. Thus, the embedding

$$A \to A *_B (B \otimes (C', \kappa'))$$

is existential. The C*-probability space $(C', \kappa') = C_r^*(\mathbb{F}_\infty) * (C, \kappa)$ is both selfless and nontracial. By Gould's dichotomy [12], it is purely infinite. By Ozawa's proof that purely infinite C*-algebras are completely selfless, [25, Theorem 3], we have a *completely* existential embedding

$$(C', \kappa') \to (C', \kappa') * (\mathcal{O}_\infty, \omega).$$

Taking tensor product with B, yields a (completely) existential embedding

$$B \otimes (C', \kappa') \to (B \otimes (C', \kappa')) *_B (B \otimes \mathcal{O}_\infty, \omega).$$

Taking amalgamated reduced free product with A and using free exactness, we get an existential embedding

$$A *_B B \otimes (C', \kappa') \to A *_B (B \otimes (C', \kappa')) *_B (B \otimes \mathcal{O}_\infty, \omega).$$

So the embedding $A \to A *_B (B \otimes \mathcal{O}_\infty, \omega)$ is also existential. □

Definition 7.4. *Let us call (A, E, B) weakly selfless, if E is weakly selfless. Equivalently, if the embedding*

$$A \mapsto A *_B (B \otimes (C([-2,2]))^{*\infty})$$

is an existential embedding of C-algebras.*

We call (A, E, B) weakly Toeplitz selfless if E is weakly Toeplitz selfless. Equivalently, if the embedding

$$A \mapsto A *_B (B \otimes (\mathcal{O}_\infty, \omega))$$

is an existential embedding of C-algebras.*

Remark 7.5. Hirshberg and Phillips show in [17, Corollary 3.5] that

$$C([-2,2])^{*\infty} \cong \mathcal{Z}^{*\infty},$$

where the free products are taken with respect to the semicircular state on $C([-2,2])$ and the unique tracial state on $\mathcal{Z}$. Thus, $\mathcal{Z}^{*\infty}$ may be used in place of $C([-2,2])^{*\infty}$ in the preceding definition. Alternatively, without appealing to this isomorphism, we can deduce from [28, Proposition 6.3.1] that $C([-2,2])^{*\infty}$ and $\mathcal{Z}^{*\infty}$ embed into

each other. This is enough to show that either algebra may be used in the definition of weak selflessness.

By Theorem 5.3, the expectation of a weakly selfless (A, E, B) is a.s.a. inner. In the following theorem we give a stronger version of this property in this new setting:

Theorem 7.6. *Let (A, E, B) be a weakly selfless C*-probability space. Then there exists a unitary $w \in B' \cap A^{\mathcal{U}}$ such that*

$$\left\| E(a) - \frac{1}{n}\sum_{k=1}^{n} (w^k)^* a w^k \right\| \leq \left(\frac{2}{n} + \frac{2}{\sqrt{n}}\right)\|a\|$$

for all $a \in A$ and $n \in \mathbb{N}$. Consequently, $E(a) \in \overline{\mathrm{co}\{u^* a u : u \in B' \cap U(A^{\mathcal{U}})\}}$ for all $a \in A$.

Proof. We work in $A *_B (B \otimes C(\mathbb{T}))$, and continue to denote the free product expectation by E. Let $u \in C(\mathbb{T})$ be a Haar unitary generating $C(\mathbb{T})$. Let

$$A_k = (u^k)^* A u^k \qquad k \in \mathbb{Z}.$$

We note that, as in the scalar case, the C*-subalgebras A_k are free with amalgamation over B, and the restriction of the free product expectation to $C^*(A_1, A_2, \ldots)$ induces a faithful GNS representation, so that

$$C^*((A_k)_{k\in\mathbb{Z}}) \cong (*_B)_{k\in\mathbb{Z}} A_k.$$

See the proof of [6, Lemma 4.1].

Now let $a \in A = A_0$, and set $x_k = E(a) - (u^k)^* a u^k \in A_k$ for $k \geq 0$. Note that $E(x_k) = 0$ for all k. By Voiculescu's inequality [19, Proposition 7.4],

$$\left\| \sum_{k=1}^{n} x_k \right\| \leq \sup_k \|x_k\| + \left\| \sum_{k=1}^{n} E(x_k^* x_k) \right\|^{\frac{1}{2}} + \left\| \sum_{k=1}^{n} E(x_k x_k^*) \right\|^{\frac{1}{2}}.$$

We have $\|x_k\| \leq 2\|a\|$ and $E(x_k^* x_k) = E(a^* a) - E(a)^* E(a)$, so $\|E(x_k^* x_k)\| \leq \|a\|^2$ and similarly $\|E(x_k x_k^*)\| \leq \|a\|^2$. We thus get

$$\left\| nE(a) - \sum_{k=1}^{n} (u^k)^* a u^k \right\| \leq 2\|a\| + 2n^{\frac{1}{2}}\|a\|.$$

Divide by n, apply $\sigma\colon A *_B (B \otimes C(\mathbb{T})) \to A^{\mathcal{U}}$ and set $w = \sigma(u)$ to get the result. □

If (A, ρ) is selfless in the scalar case, the Dixmier property of ρ readily implies that A is simple and can have at most one trace [29, Theorem 3.1]. In the relative case, we have the following:

Corollary 7.7. *If (A, E, B) is weakly selfless, then every bounded trace on A factors through B via E and for each nonzero ideal $I \subseteq A$, we have that $E(I) = I \cap B \neq 0$.*

Proof. Let τ be a bounded trace on A and $\tau^{\mathcal{U}}$ its ultrapower in $A^{\mathcal{U}}$. Then $\tau^{\mathcal{U}}$ remains constant throughout the set $\overline{\mathrm{co}\{uau^* : u \in U(A^{\mathcal{U}})\}}^{\|\cdot\|}$. Thus, by the Dixmier averaging property of E, we have $\tau(E(a)) = \tau(a)$ for all $a \in A$.

Let I be a nonzero ideal of A. By the Dixmier averaging property of E, we have $E(I) \subseteq I$, and so $E(I) = I \cap B$. We cannot have $E(I) = 0$, since we have assumed that E induces a faithful GNS. Thus, $I \cap B \neq 0$. □

Permanence properties:

Theorem 7.8. *(i) If (A, E, B) is selfless, then so is $(A *_B A', E *_B E', B)$ for an arbitrary (A', E', B).*
(ii) If $p \in B$ is a projection, then $(pAp, E|_{pAp}, pBp)$ is selfless.
(iii) Given (A, E, B) and a conditional expectation $F\colon B \to C$, if (A, FE, C) is selfless and F has finite Watatani index, then (A, E, B) is selfless.
(iv) If (A, E, B) is selfless and $F\colon B \to C$ is an a.s.a. inner expectation, then (A, FE, C) is selfless.
(v) If (A_1, E_1, B_1) is selfless, A_2 is exact, and $E_2\colon A_2 \to B_2$ is a.s.a. inner, then $(A_1 \otimes A_2, E_1 \otimes E_2, B_1 \otimes B_2)$ is selfless.

Proof. (i) This follows from "free exactness".

(ii) This follows from the corners version of Blanchard-Dykema.

(iii) By finite Watatani index we mean the existence of a quasibasis $\{x_i, y_i : i = 1, \dots, n\}$ in B such that

$$b = \sum_{i=1}^{n} y_i F(x_i b) = \sum_{i=1}^{n} F(b y_i) x_i$$

for all $b \in B$. We note that the quasibasis can always be chosen of the form $\{x_i, x_i^* : i = 1, \dots, n\}$ [37, Lemma 2.1.6]. Let us assume that it has this form. The element $e = \sum_{i=1}^{n} x_i x_i^*$ is the a central invertible element in C called the index of F. Set $x_i' = e^{-1/4} x_i$ for $i = 1, \dots, n$. Then it is not hard to check that

$$b = \sum_{i,j} x_i' F((x_i')^* b x_j')(x_j')^*,$$

for all $b \in B$ (see [14, Example 5.2.2]). Then

$$E(a) = \sum_{i,j} x_i' FE((x_i')^* a x_j')(x_j')^*.$$

for all $a \in A$. Since FE is selfless, it follows by Theorem 5.4 that E is selfless.

(iv) Since E is weakly selfless, Theorem 5.3 implies that E is a.s.a. inner; hence FE is a.s.a. inner. We can thus apply Theorem 5.4 with $\phi = E$ and $\psi = FE$.

(v) This follows from Theorem 5.4. □

Corollary 7.9. *If (A_1, ρ_1) is selfless and (A_2, ρ_2) is exact, simple, with unique trace ρ_2, then $(A_1 \otimes A_2, \rho_1 \otimes \rho_2)$ is selfless.*

Proof. If A_2 is simple with unique tracial state ρ_2, then A_2 has the Dixmier property by the theorem of Haagerup and Zsidó [13]. It follows that $x \mapsto \rho_2(x) 1_{A_2}$ is approximately inner. It is easy to see from this that $\rho_2 1_{A_2}$ is a.s.a. inner (using that it ranges $\mathbb{C}1$). We can thus apply Theorem 7.8. □

Theorem 7.10. *Let (A, E, B) be a C*-probability space with B finite dimensional and E faithful. Then (A, E, B) is selfless if and only if*

$$(A, E, B) \cong (A_1, E_1, B_1) \oplus \cdots \oplus (A_m, E_m, B_m)$$

where each A_i is selfless in the scalar sense relative to a state ρ_i, and if ρ_i is tracial then it is preserved by E_i.

Proof. Suppose that (A, E, B) is selfless. Let us first sort out the direct sum decomposition of A. Let I be an ideal of A. By Corollary 7.7, we have that $E(I) = I \cap B$

is an ideal of B. Hence, it has the form $e_I B$ for some central projection $e_I \in B$. If $a \in I$, then

$$E((1-e_I)aa^*(1-e_I)) = (1-e_I)E(aa^*)(1-e_I) = 0.$$

By the faithfulness of E we deduce that $e_I a = a$. Thus, $I = e_I A$. We deduce from this that A decomposes into a direct sum of simple C*-subalgebras $A_i = e_i A$, for $i = 1, \ldots, m$, where $e_1, \ldots, e_m$ are pairwise orthogonal central projections of B. We thus obtain the direct sum decomposition of the theorem with $E_i = E|_{A_i}$ and $B_i = e_i B$. Note that, by the preservation of selflessness when passing to corners of B (Theorem 7.8), each (A_i, E_i, B_i) is selfless.

Let us sort out next the structure of the simple summands. Fix i, and pick a minimal projection $p \in B_i$. This way $pB_ip \cong \mathbb{C}$. By Theorem 7.8, $(pA_ip, E|_{pA_ip}, \mathbb{C})$ is selfless. That is, pA_ip is selfless in the scalar sense relative to the state $\rho_{p,i} = E|_{pA_ip}$. Since A_i is simple, the projection p generates A_i as an ideal. Thus, by Brown's theorem, A_i is isomorphic to a corner of a matrix algebra: $A_i \cong qM_N(pA_ip)q$. Since selflessness is preserved under matrix amplifications and corners ([29], [12]), A_i is selfless. If A_i is tracial, then its trace must be preserved by E_i, by Corollary 7.7.

Suppose conversely that (A, E, B) admits the direct sum decomposition of the theorem. Given C*-probability spaces (A, E, B), (C, F, B), (A', E', B'), (C', F', B'), we have a canonical isomorphism

$$(A \oplus A') *_{B\oplus B'} (C \oplus C') \cong (A *_B C) \oplus (A' *_{B'} C').$$

From this we deduce that it suffices to show that each summand is selfless.

Assume that A is selfless in the scalar sense, that $E\colon A \to B$ is a faithful conditional expectation onto a finite dimensional subalgebra, and that if A is tracial, then E preserves the trace. If A is not tracial, then it is purely infinite, and so it is selfless relative to any state. In this case, choose a faithful trace $\tau\colon B \to \mathbb{C}$ and set $\rho = \tau \circ E$. Then in both the tracial or the non-tracial case, we can arrange that (A, ρ) is selfless where ρ is faithful, preserved by E, and inducing a trace on B. Then $(B, \rho|_B)$ has finite Watatani index, as B is finite dimensional [37]. Thus, (A, E, B) is selfless by Theorem 7.8. □

8. Interactions of selflessness with nuclearity

The definition of relative nuclearity below appears in Hasegawa [14] (applied to conditional expectations).

Definition 8.1. *Let $\phi \in CP(A)$. Call ϕ real relatively nuclear if id_A is a point-norm limit of cp maps of the form*

$$a \mapsto \sum_{k=1}^{m} \sum_{i,j=1}^{n_k} x_{k,i}\phi(x_{k,i}^* a x_{k,j})x_{k,j}^*, \qquad a \in A,$$

where $x_{k,i} \in A$. Call ϕ relatively nuclear if the identity id_A is a point-norm limit of cp maps of the form

$$a \mapsto \sum_{k=1}^{m} \sum_{i,j=1}^{n_k} y_{k,i}\phi(x_{k,i}^* a x_{k,j})y_{k,j}^*, \qquad a \in A,$$

where $x_{k,i}, y_{k,i} \in A$.

Remark 8.2. It is not difficult to show that ϕ is relatively nuclear if and only if we have the weak containment of C*-correspondences $A \precsim A \otimes_\phi A$, and it is real relatively nuclear if and only if we have the weak containment of C*-correspondences with real structures $(A, \mathrm{std}) \precsim (A \otimes_\phi A, \mathrm{std})$.

Theorem 8.3. *Let $E\colon A \to B$ be a unital conditional expectation. Let $\psi \in CP(B)$.*

(i) If $(A, \psi\circ E)$ is weakly selfless and ψ is real relatively nuclear, then (A, E, B) is weakly selfless.

(ii) If $(A, \psi \circ E)$ is weakly Toeplitz selfless and ψ is relatively nuclear, then (A, E, B) is weakly Toeplitz selfless.

Proof. (i) Choose a net of maps $(\psi_\lambda)_\lambda \in CP(B)$ as in the definition of real relative nuclearity applied to ψ, converging to the identity of B in the point-norm topology. From the fact that $\psi \circ E$ is weakly selfless, we deduce that $\psi_\lambda \circ E$ is weakly selfless by Theorem 5.4 (i). Hence $E = \lim_\lambda \psi_\lambda \circ E$ is weakly selfless by Theorem 5.4.

(ii) We argue as in (i), using the Toeplitz version of the cp-map permanence theorem. The more general form of the cp maps appearing in the definition of relative nuclearity is handled by Theorem 5.11. Thus $\psi_\lambda \circ E$ is weakly Toeplitz selfless for each λ, and the point-norm limit argument gives weak Toeplitz selflessness. □

Corollary 8.4. *Let A be a separable unital C*-algebra and let $\phi \in CP(A)$.*

(i) If ϕ is weakly selfless and real relatively nuclear, then $A \otimes \mathcal{Z} \cong A$.

(ii) If ϕ is weakly Toeplitz selfless and relatively nuclear, then $A \otimes \mathcal{O}_\infty \cong A$.

Proof. For part (i), apply Theorem 8.3 with $B = A$, $E = \mathrm{id}_A$, and $\psi = \phi$. Since $\phi = \phi \circ \mathrm{id}_A$ is weakly selfless and ϕ is real relatively nuclear, the theorem shows that (A, id_A, A) is weakly selfless. Consequently, there is a *-homomorphism

$$A \otimes \mathcal{Z}^{*\infty} \cong A *_A (A \otimes \mathcal{Z}^{*\infty}) \longrightarrow A^{\mathcal{U}}$$

whose restriction to A is the diagonal embedding. Restricting this homomorphism to $1_A \otimes \mathcal{Z}$ gives a unital embedding $\mathcal{Z} \to A' \cap A^{\mathcal{U}}$. Since A is separable, $A \otimes \mathcal{Z} \cong A$ [34, Theorem 2.2].

To prove (ii), apply Theorem 8.3 (ii) with $B = A$, $E = \mathrm{id}_A$, and $\psi = \phi$. This yields a unital embedding $\mathcal{O}_\infty \to A' \cap A^{\mathcal{U}}$. Since A is separable, $A \otimes \mathcal{O}_\infty \cong A$ [30]. □

Remark 8.5. Expanding the proof of (i) some more, we can show this: if A is exact, then (A, ϕ) is both selfless and real relatively nuclear if and only if ϕ is a.s.a. inner and the first factor embedding $(A, \phi) \to (A \otimes \mathcal{Z}, \phi \otimes \tau)$ is existential. For separable C*-algebras, a C*-algebraic existential embedding into $A \otimes \mathcal{Z}$ is equivalent to $\mathcal{Z}$-stability, but the standard proof of this fact does not directly produce an isomorphism $(A, \phi) \cong (A \otimes \mathcal{Z}, \phi \otimes \tau)$ assuming that the first factor embedding $(A, \phi) \to (A \otimes \mathcal{Z}, \phi \otimes \tau)$ is existential and A separable; we instead get a sequence of C*-algebraic isomorphisms that asymptotically intertwine ϕ and $\phi \otimes \tau$.

Corollary 8.6. *[Kirchberg's $\mathcal{O}_\infty$-stability theorem] If A is separable, purely infinite, simple, unital, and nuclear, then $A \cong A \otimes \mathcal{O}_\infty$.*

Proof. Choose any state ρ on A. By Ozawa's theorem (or rather, its proof), (A, ρ) is Toeplitz selfless [25, Theorem 3]. Since A is nuclear and ρ induces a faithful GNS representation, $(A, \rho, \mathbb{C})$ is relatively nuclear [14, Example 5.1.1]. By the previous corollary, $A \cong A \otimes \mathcal{O}_\infty$. □

A tracial C*-probability space (A, ρ) is called MF if it admits a trace preserving embedding into $(\mathcal{Q}^{\mathcal{U}}, \tau^{\mathcal{U}})$, where $\mathcal{Q} = \bigotimes_{n=1}^{\infty} M_n(\mathbb{C})$. In the theorem below we apply Theorem 8.3 to show that certain reduced amalgamated free products of MF C*-probability spaces are again MF.

Theorem 8.7. *Let (A, ρ) be an MF C*-probability space. Let $E\colon A \to B$ be trace preserving expectation onto a real nuclear C*-subalgebra B. Then*

$$(A *_B A, \rho \circ (E *_B E))$$

is MF.

Proof. We first reduce the proof to the case that (A, ρ) is selfless. Consider

$$(A', \rho') = (A, \rho) * (\mathcal{Q}, \tau).$$

Then (A', ρ') is selfless and MF [29]. The expectation $E' = E * \tau\colon A' \to B$ extends E and preserves ρ'. By Blanchard-Dykema, $A *_B A$ embeds in an expectation preserving way into $A' *_B A'$, so it suffices to show that $A' *_B A'$ is MF. Renaming (A', ρ') as (A, ρ), let us henceforth assume that (A, ρ) is selfless.

Applying Theorem 8.3 with $\psi = \rho|_B$, we deduce that (A, E, B) is weakly selfless. In particular, $C = A *_B (B \otimes C_r^*(\mathbb{F}_\infty))$ embeds in $A^{\mathcal{U}}$. To see that this embedding is trace preserving, it will suffice to show that C has a unique trace. Let $F\colon C \to B$ be the amalgamated free product conditional expectation onto B. Notice that (C, F, B) is selfless, by Theorem 7.8 (i), (v). Thus, every trace on C factors through F. But if θ is a tracial state on C, then its restriction to A must agree with ρ, by selflessness of (A, ρ). Hence,

$$\theta = \rho|_B \circ F.$$

This shows that C has a unique trace, whence its embedding in $A^{\mathcal{U}}$ is trace preserving.

We obtain a trace-preserving embedding of $A *_B A$ into C by choosing a Haar unitary $u \in C_r^*(\mathbb{F}_\infty)$ and mapping the first copy of A identically and the second copy to $(1 \otimes u)A(1 \otimes u)^*$. Thus, $A *_B A$ embeds in $A^{\mathcal{U}}$ in a trace preserving way. Since A is MF, we conclude that $A *_B A$ is MF. □

A discrete group G is called MF if $(C_r^*(G), \tau)$ is MF, where τ is the canonical trace.

Corollary 8.8. *Let G be an MF group with amenable subgroup H. Then $G *_H G$ is MF.*

Proof. We have

$$C_r^*(G *_H G) \cong C_r^*(G) *_{C_r^*(H)} C_r^*(G)$$

[36, Chapter 4]. Moreover, if $H \subseteq G$ is amenable, then $C_r^*(H)$ is real nuclear [5, Theorem 2.6.8]. We can thus apply the previous theorem. □

9. The PHP property and crossed products

In [25] Ozawa introduced the PHP property for groups. We recall its definition here (stated slightly differently):

Definition 9.1. *The discrete group G has the PHP property if, for every finite set $\Lambda \subseteq G \setminus \{e\}$, every $\delta > 0$, and every $N \in \mathbb{N}$, there exist $n \geq N$ and triples $(t_i, \Gamma_i, \Gamma_i^+)$, $i = 1, \ldots, n$, with $t_i \in G$ and $\Gamma_i^+ \subseteq \Gamma_i \subseteq G$, satisfying the following conditions:*

P1: The sets

$$\Gamma_i^+,\ t_j^{-1}(G\setminus\Gamma_j),\quad i,j=1,\dots,n$$

are pairwise disjoint.

P2: With

$$\Gamma=\bigcup_i\Gamma_i^+\cup\bigcup_i t_i^{-1}(G\setminus\Gamma_i),$$

we have $g\Gamma\cap\Gamma=\varnothing$*, for all* $g\in\Lambda$*,* $e\notin\Gamma$*, and* $g^{-1}\notin\Gamma$ *for all* $g\in\Lambda$*.*

P3: For each $g\in G$*, the cardinalities of the sets*

$$\{i: g\in\Gamma_i\setminus\Gamma_i^+\},\qquad \{i: g\in t_i^{-1}(\Gamma_i\setminus\Gamma_i^+)\}$$

are at most $\delta\sqrt{n}$*.*

By [25, Theorem 14], if G has the PHP property then $C_r^*(G)$ is completely selfless, whence selfless. The class of groups that have the PHP property is closed under direct products [25, Section 8] and contains all acylindrically hyperbolic groups with trivial finite radical [25, Proposition 15].

Let G be a discrete group acting on a unital C*-algebra B: $g\mapsto\alpha_g\in\mathrm{Aut}(B)$. We recall the construction of the reduced crossed product $B\rtimes_r G$ and of the expectation $E\colon B\rtimes_r G\to B$, and along the way introduce the notation that we will use below.

Assume that B is embedded unitally in $B(H)$, for a Hilbert space H. Let $\ell^2(G,H)$ be the Hilbert space of square summable functions $\xi\colon G\to H$. Let B and G act on $\ell^2(G,H)$ as follows:

$$(b\xi)(g)=\alpha_{g^{-1}}(b)\xi(g),\qquad (u_h\xi)(g)=\xi(h^{-1}g),$$

for $b\in B$, $g,h\in G$, and $\xi\in\ell^2(G,H)$. Then $B\rtimes_r G=C^*(B,u_g:g\in G)$. We note that $u_gb=\alpha_g(b)u_g$, from which it follows that finite sums of the form

$$\sum_g b_gu_g$$

form a dense *-subalgebra of $B\rtimes_r G$.

Let $V_e\colon H\to\ell^2(G,H)$ be the embedding of H into the summand at $g=e$, i.e., $V_e(v)=v\delta_e$ for $v\in H$. Note that $V_eb=bV_e$ for $b\in B$ (where b acts on H on the left-hand side and on $\ell^2(G,H)$ on the right-hand side). Define a u.c.p. map $E\colon B(\ell^2(G,H))\to B(H)$ by $E(x)=V_e^*xV_e$. Then E is a B-bimodule map and

$$E(\sum_g b_gu_g)=b_e.$$

So E defines an expectation from $B\rtimes_r G$ onto B.

Regard $\ell^\infty(G)\subset B(\ell^2(G,H))$ as multiplication operators. Set

$$A=B\rtimes_r G,\qquad C=C^*(A,\ell^\infty(G))\subseteq B(\ell^2(G,H)).$$

The finite linear span of the elements bfu_g, with $b\in B$, $f\in\ell^\infty(G)$, and $g\in G$, is a dense $*$-subalgebra of C. Since $E(bfu_g)=0$ if $g\neq e$, and $E(bf)=f(e)b$, the map E restricts to a B-valued conditional expectation on C.

Theorem 9.2. *If G is a discrete group with the PHP property acting on B by approximately inner automorphisms, then $(B\rtimes_r G,E,B)$ is selfless.*

The proof adapts Ozawa's PHP argument for reduced group C*-algebras [25, Theorem 14] to the relative setting, with approximate innerness used to absorb the coefficients from B.

Proof. Set $A = B \rtimes_r G$ and $C = C^*(A, \ell^\infty(G))$, and regard E as a B-valued conditional expectation on C. We will construct for each finite set $F \subseteq A$ and $\epsilon > 0$, an operator $T \in C$ such that

(1) $T^*aT \approx_\epsilon E(a)$ for all $a \in F$,
(2) $E(aTT^*a^*) \approx_\epsilon 0$ for all $a \in F$,
(3) $T + T^* \approx_\epsilon s$ for some $s \in A$.

By Proposition 5.9, this implies that E is selfless as a cp map on A, i.e., (A, E, B) is selfless (Theorem 7.3).

Let $F \subset A$ be finite and $\epsilon > 0$. To construct T, it suffices to choose F from a set with dense linear span. So let us assume that

$$F = \{b_g u_g : g \in \Lambda\} \cup S,$$

where $\Lambda \subseteq G \setminus \{e\}$, $b_g \in B$, and $1 \in S \subset B$.

Let $\delta > 0$ be small (how small to be made precise below). By the PHP property of G applied to the finite set Λ and to δ, there exist n and triples $(t_i, \Gamma_i, \Gamma_i^+)$, $i = 1, 2, \ldots, n$, with $t_i \in G$ and $\Gamma_i^+ \subseteq \Gamma_i \subseteq G$, satisfying P1–P3 from Definition 9.1 and $\|b\|/\sqrt{n} < \delta$ for all $b \in S$.

From the triples $(t_i, \Gamma_i, \Gamma_i^+)$ define triples (u_i, P_i, P_i^+), where $u_i = u_{t_i} \in A$ is a unitary, and $P_i, P_i^+ \in \ell^\infty(G) \subseteq B(\ell^2(G))$ are the projections

$$P_i = P_{\Gamma_i}, \quad P_i^+ = P_{\Gamma_i^+}.$$

We regard u_i, P_i, P_i^+ as operators in C. Note that the projections P_i, P_i^+ commute with B, since elements of B act fiberwise on vectors in $\ell^2(G, H)$. Set

$$P = \sum_{i=1}^n P_i^+ + \sum_{i=1}^n u_i^*(1 - P_i)u_i.$$

From the properties of the triples $(t_i, \Gamma_i, \Gamma_i^+)$ we obtain that

- the $2n$ projections $P_i^+, u_i^*(1 - P_i)u_i$, $i = 1, \ldots, n$, are pairwise orthogonal (whence, P is a projection);
- $Pu_gP = 0$ for all $g \in \Lambda$, $PV_e = 0$, and $Pu_{g^{-1}}V_e = 0$ for all $g \in \Lambda$.
- $$\Big\| \sum_{i=1}^n (P_i - P_i^+) \Big\| \leq \delta\sqrt{n}, \qquad \Big\| \sum_{i=1}^n u_i^*(P_i - P_i^+)u_i \Big\| \leq \delta\sqrt{n}.$$

Since $\alpha_{t_i^{-1}}$ is approximately inner, we may choose for each $i = 1, \ldots, n$ a unitary $w_i \in U(B)$ such that

$$\|w_i b w_i^* - \alpha_{t_i^{-1}}(b)\| < \delta,$$

for all $b \in S$.

Define

$$T = \frac{1}{\sqrt{2n}} \sum_{i=1}^n (P_i^+ u_i w_i + w_i^* u_i^*(1 - P_i)) \in C.$$

The $2n$ summands in the definition of T have pairwise orthogonal ranges: the range of $P_i^+ u_i w_i$ is contained in P_i^+, while the range of $w_i^* u_i^*(1 - P_i)$ is contained in $u_i^*(1 - P_i)u_i$. Hence the cross terms in T^*T vanish, and

$$T^*T = \frac{1}{2n} \sum_{i=1}^n \Big(w_i^* u_i^* P_i^+ u_i w_i + (1 - P_i)u_i w_i w_i^* u_i^*(1 - P_i) \Big) \leq 1.$$

In particular, $\|T\| \leq 1$.

Let us show that T satisfies the three properties (1)-(3) listed at the start of the proof for a sufficiently small δ.

Proof of (1). Let $x = b_g u_g$, with $g \in \Lambda$. As remarked above, P_i and P_i^+ commute with $b_g \in B$ for all i. Thus

$$PxP = Pb_g u_g P = b_g P u_g P = 0.$$

The range of $P_i^+ u_i w_i$ is contained in the range of P_i^+, whereas the range of $w_i^* u_i^* (1 - P_i)$ is contained in the range of $u_i^*(1 - P_i)u_i$. We deduce from this that $T = PT$. Hence,

$$T^* x T = T^* P x P T = 0.$$

Now let $b \in S$. A direct computation shows that

$$T^* b T = \frac{1}{2n} \sum_{i=1}^{n} \Big(w_i^* \alpha_{t_i^{-1}}(b) w_i \, u_i^* P_i^+ u_i + \alpha_{t_i}(w_i b w_i^*)(1 - P_i) \Big).$$

By our choice of the unitaries w_i, $\|w_i^* \alpha_{t_i^{-1}}(b) w_i - b\| < \delta$ and $\|\alpha_{t_i}(w_i b w_i^*) - b\| < \delta$. Applying these estimates on the right hand side of the expression for $T^* b T$, we get

$$\Big\| T^* b T - \frac{1}{2n} \sum_{i=1}^{n} (b u_i^* P_i^+ u_i + b(1 - P_i)) \Big\| < \delta.$$

Thus

$$\begin{aligned} \|T^* b T - b\| &\leq \delta + \frac{\|b\|}{2n} \Big\| \sum_i (u_i^*(1 - P_i^+)u_i + P_i) \Big\| \\ &\leq \delta + \frac{\|b\|}{2n} \|P\| + \frac{\|b\|}{2n} \Big\| \sum_i u_i^*(P_i - P_i^+)u_i \Big\| + \frac{\|b\|}{2n} \Big\| \sum_i (P_i - P_i^+) \Big\| \\ &\leq 3\delta. \end{aligned}$$

Choosing δ sufficiently small and n sufficiently large gives (1).

Proof of (2). Since $PV_e = 0$, we have $E(P) = V_e^* P V_e = 0$. Let $b \in B$. Since T is a contraction and $PT = T$, we have $TT^* \leq P$. Hence,

$$0 \leq E(bTT^*b^*) \leq bE(P)b^* = 0.$$

Now let $x = b_g u_g$, with $g \in \Lambda$. Since $TT^* \leq P$,

$$xTT^*x^* \leq b_g u_g P u_{g^{-1}} b_g^*.$$

Applying E we get

$$0 \leq E(xTT^*x^*) \leq b_g E(u_g P u_{g^{-1}}) b_g^*.$$

Since $P u_{g^{-1}} V_e = 0$,

$$E(u_g P u_{g^{-1}}) = V_e^* u_g P u_{g^{-1}} V_e = 0.$$

Thus, $E(xTT^*x^*) = 0$. This proves (2).

Proof of (3). Set

$$s = \frac{1}{\sqrt{2n}} \sum_{i=1}^{n} (u_i w_i + w_i^* u_i^*) \in B \rtimes_r G.$$

Since w_i commutes with P_i and P_i^+,

$$T + T^* - s = -\frac{1}{\sqrt{2n}} \sum_{i=1}^{n} ((P_i - P_i^+)u_i w_i + w_i^* u_i^* (P_i - P_i^+)).$$

By the operator Cauchy-Schwarz inequality,

$$\Big\| \sum_{i=1}^{n} (P_i - P_i^+)u_i w_i \Big\| \leq \Big\| \sum_{i=1}^{n} (P_i - P_i^+) \Big\|^{\frac{1}{2}} \Big\| \sum_{i=1}^{n} u_i^* (P_i - P_i^+)u_i \Big\|^{\frac{1}{2}} \leq \delta\sqrt{n}.$$

Hence

$$\|T + T^* - s\| \leq \sqrt{2}\delta.$$

Thus, choosing $\sqrt{2}\delta < \epsilon$ proves (3). □

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